From the three-particle Green's function to the parquet equations, to ADC, and beyond

The Faddeev-decomposition of the three-particle vertex is used to derive a Hermitian effective Hamiltonian of Faddeev-ADC(3) type. Besides the Parquet approximation and the use of a Hartree–Fock Green's function, the derivation is based on two further approximations: couplings between different correlation channels as well as overlaps between different time-orderings are neglected. Faddeev-ADC(3) contains FLEX (first-order resolvent), ADC(2)-X (TDA limit), ADC(3) (TDA limit + first-order backward couplings) as limiting cases. Several self-energy schemes that are derived in Hedin's scheme or in the pp channel as well as PSD-completions (double counting of second-order terms aside) are recovered as a limiting case of Faddeev-ADC(3), and they are reviewed. Numerical results show that Faddeev-ADC(3), as well as all other methods that explicitly include triplet backward couplings to all orders come with very large errors. Switching off the triplets largely overcomes these issues, but only the performance of GW is recovered. Replacing the ring- and ladder-CCD amplitudes of Faddeev-ADC(3) with CCD ones, one recovers essentially the performance of CCSD. Coupling correlation channels seems to be decisive for accurate charged excitations.

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I. INTRODUCTION

Via the three-particle Green's function G_3, two routes to the one-particle self-energy are connected: the parquet equations for the two-particle vertex and the algebraic-diagrammatic construction (ADC). Starting from the Martin–Schwinger hierarchy (Sec. II A), we express the correlation self-energy in terms of a 6-point response function R, which obeys a Dyson equation with a three-particle kernel (Sec. II B). Dropping its 3-body irreducible part leaves the Faddeev kernel,[1–3] in which two of the three lines scatter through a two-particle vertex while the third line is a spectator. In the Faddeev form, this vertex is the full parquet vertex F (Sec. II C).[4]

Truncated at fixed order in v, the Faddeev series gives GFn, and Schirmer's matching gives ADC(n) (App. D). The connection becomes useful once the ladders in the pairs are resummed beyond the Tamm–Dancoff ladders contained in ADC (Sec. III). With the RPA ladders of the first-order Faddeev kernel in every pair, and without overlaps between the ladders of different pairs, a Hermitian effective Hamiltonian of the size of ADC(3) is obtained, that describes the coupling of a single particle to pp and eh phonons. This is essentially the Faddeev random-phase approximation (FRPA) of Barbieri and co-workers.[5–8] In FRPA, the couplings are those of ADC(3), $U^{(1)}+U^{(2)}$, added by

hand;[8] here they follow from the backward amplitudes of the phonons, the ring-CCD (eh) and ladder-CCD (pp) amplitudes of the channels.[9] The hierarchy is ADC(3) ⊂ FRPA ⊂ parquet for the dynamic self-energy. At fourth order, the crossed channels, the overlaps and the line dressing are added in the 3p2h space.

Three approximations are made by construction. The parquet approximation, $\Lambda = \Lambda^{(1)}$, keeps F exact through third order, since $\Lambda = \Lambda^{(1)} + O(v^4)$, Eq. (A4). With a static HF reference in all lines of the vertex, every diagram of the vertex is a product of instantaneous interactions and G_{HF} lines, whose frequency integrals turn into resolvents of configuration spaces. This makes the effective-Hamiltonian form possible. Finally, the Faddeev approximation affects the self-energy from fifth order on. The orders through which the resulting self-energies are exact are collected in Table I; the constructions derived in these notes are exact through fourth order at most.

TABLE I. Order through which the self-energy is exact, and the highest ADC(n) contained, for different approximations. Λ: fully irreducible two-particle vertex; Ξ_{3B}: 3-body irreducible kernel; SDE: Schwinger–Dyson equation. The second row is the construction of these notes. The first three rows use the parquet approximation, $\Lambda = \Lambda^{(1)}$, Eq. (A4).

Approximation	Eqs.	exact through	contains
Parquet, SDE	(8), (A7)	4th	ADC(4)
Parquet F, Faddeev	(19), (21)	4th	ADC(4)
Parquet $F + \Xi_{3B}^{(3)}$	(16), (13)	5th	ADC(5)
$\Lambda^{(\leq 4)}$, SDE	(8), (A7)	5th	ADC(5)
$\Lambda^{(\leq 4)} + \Xi_{3B}^{(\leq 4)}$	(16), (13)	6th	ADC(6)

II. FADDEEV APPROACH TO THE SELF-ENERGY

A. Martin–Schwinger Equations

Let us define the n-particle Green's function as

$$G_n(1\dots n;1'\dots n') = (-\mathrm{i})^n\langle \mathcal{T}_C\, \hat\psi(1)\cdots\hat\psi(n) \times \hat\psi^\dagger(n')\cdots\hat\psi^\dagger(1')\rangle\,, \tag{1}$$

with $\hat\psi$ in the Heisenberg picture. G_n might be expressed in terms of G_{n-1} and G_{n+1} through the Martin–Schwinger hierarchy,[10]

$$\begin{aligned}&[\mathrm{i}\partial_{t_1} - h_0(1)]\, G_n(1\dots n;1'\dots n')\\ &= \sum_{j=1}^{n}(-1)^{j-1}\delta(1j')\, G_{n-1}(2\dots n;1'\dots\widehat{j'}\dots n')\\ &\quad + (-1)^n\,\mathrm{i}\, v(1\bar2;\bar3\bar4)\, G_{n+1}(\bar3\bar4\,2\dots n;1'\dots n'\bar2^+)\,.\end{aligned} \tag{2}$$

We define

$$G_0^{-1}(11') = \left[\mathrm{i}\frac{\partial}{\partial t_1} - h_0(1)\right]\delta(11')\,, \tag{3}$$

and its inverse via

$$G_0^{-1}(1\bar1)\, G_0(\bar11') = \delta(11')\,. \tag{4}$$

Using both relations, G_1 through G_3 become

$$\begin{aligned}G(11') =& G_0(11')\\ &- \mathrm{i}\, G_0(1\bar3)\, v(\bar3\bar4;\bar5\bar6)\, G_2(\bar5\bar6;1'\bar4^+)\,,\end{aligned} \tag{5}$$

$$\begin{aligned}G_2(12;1'2') =& G_0(11')G(22') - G_0(12')G(21')\\ &+ \mathrm{i}\, G_0(1\bar3)\, v(\bar3\bar4;\bar5\bar6)\\ &\times G_3(2\bar5\bar6;1'2'\bar4^+)\,,\end{aligned} \tag{6}$$

$$\begin{aligned}G_3(123;1'2'3') =& G_0(11')\, G_2(23;2'3')\\ &- G_0(12')\, G_2(23;1'3')\\ &+ G_0(13')\, G_2(23;1'2')\\ &- \mathrm{i}\, G_0(1\bar4)\, v(\bar4\bar5;\bar6\bar7)\\ &\times G_4(23\bar6\bar7;1'2'3'\bar5^+)\,,\end{aligned} \tag{7}$$

Through Eq. (5), we can define the single-particle self-energy according to

$$\Sigma(11') = -\mathrm{i}v(1\bar2;\bar3'\bar2')G_2(\bar3'\bar2';\bar3\bar2^+)G^{-1}(\bar31')\,. \tag{8}$$

B. Eliminating G_2

We can now use Eq. (6) to eliminate the dependence of Eq. (8) on G_2 in favor of G_3. G_4 might be eliminated as well, and Eq. (8) can be expressed in terms of G_3 and G_5, or, more generally, in terms of odd-order Green's functions. Alternatively, combining Eq. (6) with Eq. (7), one can express Eq. (8) in terms of G_2 and G_4, or, more generally, in terms of even-order Green's functions.

Here, we take the odd-order route. To do so, following Winter[2] and Ethofer and Schuck[3], we introduce the 6-point response function

$$\begin{aligned}R(123;1'2'3') =& G_3(123;1'2'3')\\ &- G_2(23;3'\bar4)\, G^{-1}(\bar4\bar4')\, G_2(\bar4'1;1'2')\,,\end{aligned} \tag{9}$$

so that the EOM for G_2 becomes

$$\begin{aligned}G_2(12;1'2') =& G(11')G(22') - G(12')G(21')\\ &+ \mathrm{i}\, G(1\bar3)\, v(\bar3\bar4;\bar5'\bar6')\, R(2\bar5'\bar6';1'2'\bar4^+).\end{aligned} \tag{10}$$

Plugging Eq. (10) into Eq. (8) and using Eq. (4) gives

$$\Sigma(11') = \Sigma_{\mathrm{HF}}(11') + \Sigma^c(11') \tag{11}$$

with

$$\Sigma_{\mathrm{HF}}(11') = -\,\mathrm{i}\,[v(1\bar2;1'\bar2') - v(1\bar2;\bar2'1')]\, G(\bar2'\bar2^+), \tag{12}$$

$$\begin{aligned}\Sigma^c(11') =& v(1\bar2;\bar3'\bar2')\, G(\bar3'\bar4)\, v(\bar4\bar5;\bar6'\bar7')\\ &\times R(\bar2'\bar6'\bar7';\bar3\bar2\bar5^+)G^{-1}(\bar31').\end{aligned} \tag{13}$$

The response function R obeys a Dyson equation with the 2p1h-irreducible three-particle kernel Ξ,

$$R = \mathcal{G} + \mathcal{G}\,\Xi\, R\,, \qquad R^{-1} = \mathcal{G}^{-1} - \Xi\,, \tag{14}$$

whose free propagator is the 2p1h propagator of three independent (dressed) lines,

$$\mathcal{G}(312;1'2'3') = G(33')\,[G(11')\,G(22') - G(12')\,G(21')]\ . \tag{15}$$

As in G_3, all three lines run from the second to the first triple. From here on, the first triple of R is ordered as in Eq. (15), with the line 3 of $\mathcal{G}$ in the first slot. The second triple keeps the order $(1'2'3')$, so that line 3 runs from the last slot of the second triple to the first slot of the first.

C. Faddeev Decomposition

The kernel Ξ of the Dyson equation for R is split into the Faddeev (pairwise-reducible) part and the 3-body irreducible kernel Ξ_{3B}, which is dropped from now on. In the Faddeev part, one pair of lines scatters through a two-particle vertex while the third line propagates freely.[1–3,5] Since the two lines of every pair run in the same direction, each pair scatters in its pp channel, with the pp-irreducible vertex Γ^{pp} (exact at first order, see App. A). As in Refs. 2,3, the Dyson equation for R then reads

$$R = \mathcal{G} + \mathcal{K} R\ , \qquad \mathcal{K} = \mathcal{K}_{12}[\Gamma^{pp}] + \mathcal{K}_{31}[\Gamma^{pp}] + \mathcal{K}_{32}[\Gamma^{pp}]\ , \tag{16}$$

where the pair operators act, for any 4-point Y and any χ antisymmetric in its first pair, as

$$\begin{aligned}
[\mathcal{K}_{12}[Y]\chi](312;1'2'3') &= -\frac{1}{4}[G(1\bar{1})G(2\bar{2}) - G(1\bar{2})G(2\bar{1})] \\
&\quad \times Y(\bar{1}\bar{2};\bar{1}'\bar{2}')\,\chi(3\bar{1}'\bar{2}';1'2'3'), \\
[\mathcal{K}_{31}[Y]\chi](312;1'2'3') &= G(3\bar{3})\,G(1\bar{1})\,Y(\bar{3}\bar{1};\bar{1}'\bar{3}') \\
&\quad \times \chi(\bar{3}'\bar{1}'2;1'2'3'), \\
[\mathcal{K}_{32}[Y]\chi](312;1'2'3') &= G(3\bar{3})\,G(2\bar{2})\,Y(\bar{3}\bar{2};\bar{2}'\bar{3}') \\
&\quad \times \chi(\bar{3}'1\bar{2}';1'2'3').
\end{aligned} \tag{17}$$

Eq. (16) is shown diagrammatically in Fig. 1.

Cutting the two lines of pair α separates the outgoing from the incoming legs of Y in $\mathcal{K}_\alpha[Y]$, *i.e.* all three pairs scatter in the pp channel of R. In Σ^c, Eq. (13), line 3 becomes the hole for $t_{1'} < t_1$ (block II) or the particle for $t_1 < t_{1'}$ (block I). The pair (12) then gives the ladder-type self-energy diagrams of the pp channel, and the pairs (31) and (32) give the ring-type diagrams of the eh and $\overline{\text{eh}}$ channels. From here on, we label the pairs by these channels, $\alpha \in \{\text{pp}, \text{eh}, \overline{\text{eh}}\}$ for (12), (31) and (32).

Eq. (16) is exact at first order, where all vertices are $\Gamma^{(1)}$. Beyond first order, iterating $\mathcal{K}_{\text{eh}}[\Gamma^{pp}]$ or $\mathcal{K}_{\overline{\text{eh}}}[\Gamma^{pp}]$ alone would count the rungs of the ladders in these pairs twice, since $\mathcal{G}$ is not antisymmetrized in the lines 3 and 1 (3 and 2). The pp channel of R in these pairs is the eh channel of F, whose ladder Φ^{eh} is already contained in Γ^{pp}.[8] We therefore replace Γ^{pp} in $\mathcal{K}_\alpha$ by a pair vertex Γ_α, fixed such that the pair α alone reproduces its full two-body scattering F. For the pp pair, $\Gamma_{pp} = \Gamma^{pp}$, and at first order $\Gamma_\alpha = \Gamma^{(1)}$ for all pairs. Their explicit form is not needed, since only F appears in the Faddeev form derived next.

D. Faddeev form of Eq. (13): the ADC–parquet link

1. The Faddeev Series

Eq. (16) contains the pair-irreducible vertex Γ^{pp} in all three pairs. The aim of this subsection is to rewrite R in Eq. (13) such that only the full vertex F appears in all three pairs. This is the Faddeev form of the three-body problem.[1]

Based on the Faddeev construction, we will then show that GFn is the truncation of the Faddeev series, at total order $n-2$ in v. The ADC(n) matrices then follow from Schirmer's matching,[11] in which the resolvent form of ADC(n) is imposed and matched to the self-energy order by order. ADC(3) is worked out as an illustrative case in App. D.

With the pair vertices Γ_α in place of Γ^{pp}, Eq. (16) reads

$$R = \mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[\Gamma_\alpha] R\ , \tag{18}$$

where $\mathcal{K}_\alpha[\Gamma_\alpha]$ acts only on the pair α, with the third line as a free spectator [Fig. 1(b), with the thick bars standing for Γ_α]. We collect the terms of R in which the last interaction, *i.e.* the one closest to the outgoing triple, acts on the pair α into $R_\alpha \equiv \mathcal{K}_\alpha[\Gamma_\alpha]R$. Solving $R_\alpha = \mathcal{K}_\alpha[\Gamma_\alpha](\mathcal{G} + R_\alpha + \sum_{\beta\neq\alpha} R_\beta)$ for R_α gives

$$\begin{aligned}
R &= \mathcal{G} + R_{pp} + R_{\text{eh}} + R_{\overline{\text{eh}}}\ , \\
R_\alpha &= (\mathbf{1} - \mathcal{K}_\alpha[\Gamma_\alpha])^{-1}\mathcal{K}_\alpha[\Gamma_\alpha]\Big(\mathcal{G} + \sum_{\beta\neq\alpha} R_\beta\Big)\ ,
\end{aligned} \tag{19}$$

where

$$\begin{aligned}
\mathcal{K}_\alpha[F] &= (\mathbf{1} - \mathcal{K}_\alpha[\Gamma_\alpha])^{-1}\mathcal{K}_\alpha[\Gamma_\alpha] \\
&= \mathcal{K}_\alpha[\Gamma_\alpha] + \mathcal{K}_\alpha[\Gamma_\alpha]\mathcal{K}_\alpha[F]
\end{aligned} \tag{20}$$

sums all consecutive interactions in the pair α. Acting on $\mathcal{G}$, it gives all diagrams of R in which only the pair α interacts, while the third line remains a free spectator. By the definition of Γ_α, this is the full two-body scattering of the pair. Eq. (19) are the Faddeev equations,[5,8] that, with Eq. (20) can be written compactly as

$$R_\alpha = \mathcal{K}_\alpha[F]\Big(\mathcal{G} + \sum_{\beta\neq\alpha} R_\beta\Big)\ . \tag{21}$$

Iterating Eq. (21), *i.e.* inserting the R_β on the right-hand side again and again for each channel α and inserting the result into the first line of Eq. (19) gives the Faddeev series in the form[1,5,12]

$$\begin{aligned}
R &= \mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F]\mathcal{G} + \sum_{\alpha\neq\beta} \mathcal{K}_\alpha[F]\mathcal{K}_\beta[F]\mathcal{G} \\
&\quad + \sum_{\substack{\alpha\neq\beta \\ \beta\neq\gamma}} \mathcal{K}_\alpha[F]\mathcal{K}_\beta[F]\mathcal{K}_\gamma[F]\mathcal{G} + \cdots\ ,
\end{aligned} \tag{22}$$

in which consecutive pairs are always different (Fig. 2). Eq. (22) is the starting point for two approximations. Truncated at fixed order in v, it gives GFn and, with Schirmer's matching, ADC(n). Keeping only the own-channel ladders without overlaps, but to all orders, gives Eq. (27), from which an effective Hamiltonian is derived in Sec. III.

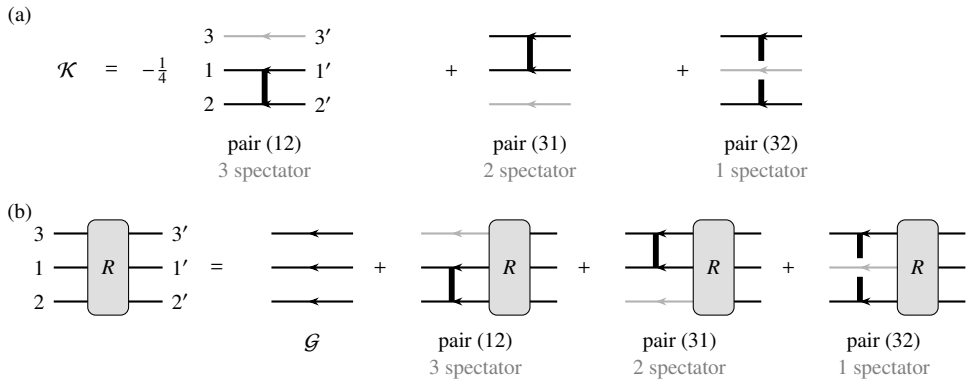


FIG. 1. (a) The Faddeev kernel, Eqs. (16) and (17). Lines run from the second triple on the right to the first triple on the left. The thick bar is the two-particle vertex Γ^{pp}: it connects the two lines of one pair, while the third line (gray) is a spectator that passes the vertex without interacting. (b) The Dyson equation for R, Eq. (16), sorted by the interacting pair. The exchange of the lines 1 and 2 in $\mathcal{G}$ is implied.

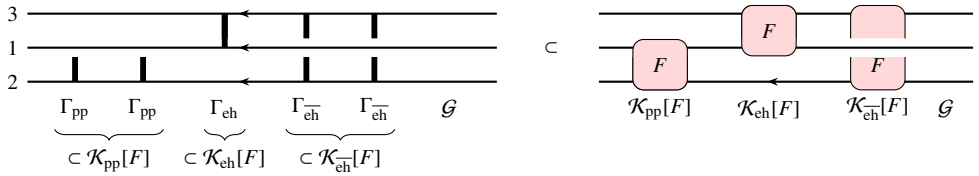


FIG. 2. Regrouping of the iterated Eq. (18) into the Faddeev series. Left: one term of the iteration, a sequence of pair vertices $\mathcal{K}_\alpha[\Gamma_\alpha]$ (thick bars). Right: each sequence of consecutive vertices in the same pair is part of the full two-body scattering $\mathcal{K}_\alpha[F]$ of that pair (shaded box), so that consecutive boxes always belong to different pairs.

2. Truncation at fixed order: ADC

To relate Eq. (22) to ADC, we expand F in powers of v. By Eqs. (A3) and (A4), Λ has no second-order term, so that

$$\begin{aligned} F^{(1)} &= \Gamma^{(1)} , \\ F^{(2)} &= \Phi^{\mathrm{pp},(2)} + \Phi^{\mathrm{eh},(2)} + \Phi^{\overline{\mathrm{eh}},(2)} . \end{aligned} \tag{23}$$

The self-energy, Eq. (13), contains two explicit interactions v. A term of order m in v of R therefore contributes to Σ at order $m+2$. ADC(n) is complete through order n, so that R is needed through order $n-2$ in v. Collecting the terms of Eq. (22) up to this order,

$$\begin{aligned} R_{\mathrm{ADC}(2)} &= \mathcal{G} , \\ R_{\mathrm{ADC}(3)} &= \mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F^{(1)}]\mathcal{G}_{\mathrm{HF}} , \\ R_{\mathrm{ADC}(4)} &= \mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F^{(1)} + F^{(2)}]\mathcal{G}_{\mathrm{HF}} \\ &\quad + \sum_{\alpha\neq\beta} \mathcal{K}_\alpha[F^{(1)}]\mathcal{K}_\beta[F^{(1)}]\mathcal{G}_{\mathrm{HF}} . \end{aligned} \tag{24}$$

Here, $\mathcal{G}$ in the first term, $G(\bar{3}'\bar{4})$ and the static part Σ_{HF} are evaluated with G expanded to the corresponding order, while in all terms with at least one F, all lines can be taken at the HF level; $\mathcal{G}_{\mathrm{HF}}$ denotes Eq. (15) with G_{HF} lines.

3. Own-Channel and No-Overlap Approximations

We now introduce the own-channel approximation to Eq. (22). For this, we approximate the BSEs Eq. (A7) in the own-channel approximation, for the special case of the first-order Faddeev kernel, so that $\Gamma^\lambda \to \Gamma^{(1)}$. The two BSEs then define two separate vertices

$$\begin{aligned} F^{\mathrm{pp}}_{\mathrm{oc}} &= \Gamma^{(1)} - \frac{1}{2}\,\Gamma^{(1)} \circ_{\mathrm{pp}} F^{\mathrm{pp}}_{\mathrm{oc}} , \\ F^{\mathrm{eh}}_{\mathrm{oc}} &= \Gamma^{(1)} + \Gamma^{(1)} \circ_{\mathrm{eh}} F^{\mathrm{eh}}_{\mathrm{oc}} . \end{aligned} \tag{25}$$

This is also done in the FLEX approximation.[13,14]

In the own-channel approximation to Eq. (22), each pair scatters only through the ladder of its own channel with the bare kernel. The own-channel approximation amounts to neglecting the channel crossings of parquet (and of coupled cluster) inside a pair.[5] Barbieri and Dickhoff[5] combined this approximation additionally with a no-overlap approximation that we denote here by $[\cdots]_{\mathrm{2p1h}}$. It neglects the time orderings in which the ladders of different pairs overlap in time, *i.e.* consecutive pairs are separated by a 2p1h cut, a subinterval spanned only by the three lines of a 2p1h state (Sec. III A). Due to this restriction, at first-order Faddeev the effective Hamiltonian only contains 2p1h (2h1p) configurations. At fourth order, the neglected terms are added perturbatively together with the 3p2h (3h2p) space. Each pair scatters with the own-channel vertex of its channel,

$$F_{\mathrm{pp}} = F^{\mathrm{pp}}_{\mathrm{oc}} , \qquad F_{\mathrm{eh}} = F_{\overline{\mathrm{eh}}} = F^{\mathrm{eh}}_{\mathrm{oc}} . \tag{26}$$

With these approximations, Eq. (22) becomes

$$\begin{aligned} R_{\rm oc} =& \mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F_\alpha]\mathcal{G} + \sum_{\alpha\neq\beta} \Big[\mathcal{K}_\alpha[F_\alpha]\mathcal{K}_\beta[F_\beta]\mathcal{G}\Big]_{\rm 2p1h} \\ &+ \sum_{\substack{\alpha\neq\beta \\ \beta\neq\gamma}} \Big[\mathcal{K}_\alpha[F_\alpha]\mathcal{K}_\beta[F_\beta]\mathcal{K}_\gamma[F_\gamma]\mathcal{G}\Big]_{\rm 2p1h} + \cdots \,. \end{aligned} \tag{27}$$

Expanded in v, Eq. (27) reproduces Eq. (24) through $R_{\rm ADC(3)}$. At the next order, it misses the crossed-channel parts of $F^{(2)}$ in each pair, Eq. (23), and the overlapping orderings of $\mathcal{K}_\alpha[F^{(1)}]\mathcal{K}_\beta[F^{(1)}]\mathcal{G}$.

III. PARQUET IN EFFECTIVE-HAMILTONIAN FORM

In this section, we insert the own-channel and no-overlap approximation, Eq. (27), into Eq. (13),

$$\begin{aligned} \Sigma^c(11') = & \, v(1\bar{2};\bar{3}'\bar{2}')\, G(\bar{3}'\bar{4})\, v(\bar{4}\bar{5};\bar{6}'\bar{7}') \\ & \times R_{\rm oc}(\bar{2}'\bar{6}'\bar{7}';\bar{3}\bar{2}\bar{5}^+)\, G^{-1}(\bar{3}1') \,. \end{aligned} \tag{28}$$

We then show that, with $G_{\rm HF}$ lines, the self-energy becomes

$$\begin{aligned} \Sigma(\omega) =& \Sigma(\infty) + \mathcal{U}^{\rm II}(\omega\mathcal{N}^{\rm II} - \mathcal{H}^{\rm II})^{-1}\mathcal{U}^{\rm II\dagger} \\ &+ \mathcal{U}^{\rm I}(\omega\mathcal{N}^{\rm I} + \mathcal{H}^{\rm I})^{-1}\mathcal{U}^{\rm I\dagger} \,, \end{aligned} \tag{29}$$

with Hermitian $\mathcal{H}$ and $\mathcal{N}$, and the static self-energy $\Sigma(\infty) = \Sigma_{\rm HF}[G]$, Eq. (12). At the level of the first-order Faddeev kernel, this is essentially the Faddeev random-phase approximation (FRPA) of Barbieri and co-workers.[5–8] The difference is that the effective Hamiltonian derived here remains Hermitian as long as its metric is positive definite, whereas the Faddeev eigenvalue problem of FRPA is non-Hermitian and contains spurious solutions.[8]

A. Resummation of the no-overlap own-channel Faddeev series

Our first goal is to evaluate the terms of Eq. (27) up to first order in $\mathcal{K}_\alpha$. It is convenient to rewrite these contributions as

$$\mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F_\alpha]\,\mathcal{G} = \sum_\alpha \Big(\mathcal{G} + \mathcal{K}_\alpha[F_\alpha]\,\mathcal{G}\Big) - 2\,\mathcal{G} \,, \tag{30}$$

which can then be evaluated channel by channel. The terms inside the sum over α are interacting single-channel three-particle propagators, for which we introduce the shorthand Q_α,

$$Q_\alpha = \mathcal{G} + \mathcal{K}_\alpha[F_\alpha]\,\mathcal{G} \,, \tag{31}$$

so that

$$\mathcal{G} + \sum_\alpha \mathcal{K}_\alpha[F_\alpha]\,\mathcal{G} = \sum_\alpha Q_\alpha - 2\mathcal{G} \,. \tag{32}$$

Truncating the Faddeev series, Eq. (27), after these terms is the one-shot FLEX approximation[13,14] (Sec. IV D). We also require that these propagators obey a three-particle Dyson-type equation,

$$Q_\alpha = \mathcal{G} + \mathcal{G}C_\alpha Q_\alpha \,, \tag{33}$$

or

$$Q_\alpha^{-1} = \mathcal{G}^{-1} - C_\alpha(\omega) \,.$$

This serves as a definition for C_α, as soon as we have specified the explicit form of Q_α.

Once we have these propagators at hand, we must evaluate the remaining terms in Eq. (28), i.e. the terms of the form

$$\mathcal{K}_\alpha[F_\alpha]\mathcal{K}_\beta[F_\beta]\mathcal{G} \,, \quad \alpha \neq \beta \,, \tag{34}$$

and so on. We can rewrite these terms using the identity and Eq. (31)

$$\begin{aligned} \mathcal{K}_\alpha[F_\alpha]\mathcal{K}_\beta[F_\beta]\mathcal{G} =& \mathcal{K}_\alpha[F_\alpha]\mathcal{G}\mathcal{G}^{-1}\mathcal{K}_\beta[F_\beta]\mathcal{G} \\ =& (Q_\alpha - \mathcal{G})\,\mathcal{G}^{-1}\Big(Q_\beta - \mathcal{G}\Big) \,, \end{aligned} \tag{35}$$

and therefore, Eq. (27) becomes

$$\begin{aligned} R_{\rm oc}(\omega) = \mathcal{G} &+ \sum_\alpha (Q_\alpha - \mathcal{G}) + \sum_{\alpha\neq\beta} (Q_\alpha - \mathcal{G})\mathcal{G}^{-1}(Q_\beta - \mathcal{G}) \\ &+ \sum_{\substack{\alpha\neq\beta \\ \beta\neq\gamma}} (Q_\alpha - \mathcal{G})\mathcal{G}^{-1}(Q_\beta - \mathcal{G})\mathcal{G}^{-1}(Q_\gamma - \mathcal{G}) + \cdots \,. \end{aligned} \tag{36}$$

Notice that this step strictly requires the no-overlap approximation, otherwise we could not just insert $\mathcal{G}\mathcal{G}^{-1}$ in each term.

To resum the series, we collect all terms whose leftmost interacting pair is α into a Faddeev component R_α. Thus,

$$R_{\rm oc} = \mathcal{G} + \sum_\alpha R_\alpha \,. \tag{37}$$

The constraint that two consecutive factors $Q_\alpha - \mathcal{G}$ must belong to different pairs implies

$$\begin{aligned} R_\alpha &= (Q_\alpha - \mathcal{G})\mathcal{G}^{-1}\left(\mathcal{G} + \sum_{\beta\neq\alpha} R_\beta\right) \\ &= (Q_\alpha - \mathcal{G})\mathcal{G}^{-1}(R_{\rm oc} - R_\alpha) \,. \end{aligned} \tag{38}$$

By the Dyson equation Eq. (34) and after some algebra, we obtain

$$(\mathbf{1} + Q_\alpha C_\alpha)R_\alpha = Q_\alpha C_\alpha R_{\rm oc} \,. \tag{39}$$

Since

$$\mathbf{1} + Q_\alpha C_\alpha = Q_\alpha(Q_\alpha^{-1} + C_\alpha) = Q_\alpha\mathcal{G}^{-1} \,, \tag{40}$$

it follows that

$$R_\alpha = \mathcal{G}C_\alpha R_{\rm oc} \,. \tag{41}$$

Summing Eq. (41) over the three pair channels and using Eq. (37) gives

$$R_{\rm oc} = \mathcal{G} + \mathcal{G}\Big(C_{\rm pp} + C_{\rm eh} + C_{\overline{\rm eh}}\Big)R_{\rm oc} \,. \tag{42}$$

Hence,

$$R_{\rm oc}(\omega) = (\mathcal{G}^{-1} - C_{\rm pp} - C_{\rm eh} - C_{\overline{\rm eh}})^{-1} \ , \tag{43}$$

or, equivalently,

$$C_\alpha = \mathcal{G}^{-1} - Q_\alpha^{-1} \ , \tag{44}$$

$$R_{\rm oc}^{-1} = Q_{\rm pp}^{-1} + Q_{\rm eh}^{-1} + Q_{\overline{\rm eh}}^{-1} - 2\mathcal{G}^{-1} \ . \tag{45}$$

Eq. (45) holds in the space $\mathcal{P}$ of the 2p1h and 2h1p configurations, where $\mathcal{G}$ lives. To insert it into Eq. (28), we still need to know how the two bare interactions of Eq. (28), at t_1 and $t_{1'}$ (the time arguments of the self-energy), connect the orbitals p and q of Σ^c_{pq} to the three-line configurations of $R_{\rm oc}$.

In the orbital basis, each line of Eq. (28) is a particle or a hole, depending on its time direction. For $R = \mathcal{G}$, all lines run between $t_{1'}$ and t_1, and only the configurations in $\mathcal{P}$ contribute. In this case, the free propagation between two equal-time three-line states is

$$\mathcal{G}(\omega) = (\omega - K)^{-1} \ , \tag{46}$$

with $\omega + \mathrm{i}0^+$ in the 2p1h and $\omega - \mathrm{i}0^+$ in the 2h1p block, and

$$K = K^{\rm II} \oplus (-K^{\rm I}) \ ,$$

$$\begin{aligned} K^{\rm II}_{abi,cdj} &= \Delta_{abi}\,\delta_{ac}\delta_{bd}\delta_{ij} \ , \\ K^{\rm I}_{ija,klb} &= -\Delta_{ija}\,\delta_{ik}\delta_{jl}\delta_{ab} \ , \end{aligned}$$

with the shorthand

$$\Delta_{xyz} = \epsilon_x + \epsilon_y - \epsilon_z \ . \tag{47}$$

The first-order couplings of the one-particle states to the three-line states are

$$U^{(1)} = (U^{\rm II,(1)},\ U^{\rm I,(1)}) \ , \tag{48}$$

with

$$\begin{aligned} U^{\rm II,(1)}_{p;abi} &= \langle pi \| ab \rangle \ , \\ U^{\rm I,(1)}_{p;ija} &= \langle pa \| ij \rangle \ . \end{aligned} \tag{49}$$

In these expressions, $a < b$ and $i < j$.

These are not all configurations that the two bare interactions of Eq. (28) can reach. If the last interaction of a pair α happens after t_1, or its first one before $t_{1'}$, the two lines of the pair arrive at the vertex in reversed time direction. For the pp pair, for instance, the vertex at t_1 then meets the three holes $(kl;i)$ instead of $(ab;i)$. Fig. 3 shows the lowest-order example. In (a), the pp rung lies between the two vertices, and all intermediate states are 2p1h configurations. In (b), the rung lies after t_1. At t_1, the bare interaction $\langle pi \| kl \rangle$ removes the electrons k and l, *i.e.* it creates the holes k and l, which end at the later rung together with the particles a and b. The pair attached at t_1 is thus the hole pair (kl), and the rung enters with the static denominator $\epsilon_k + \epsilon_l - \epsilon_a - \epsilon_b$. This is the coupling $U^{(2)}$ of ADC(3).

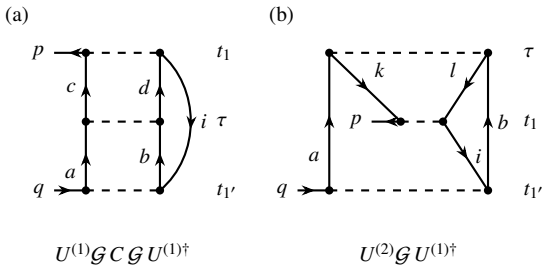


FIG. 3. Time-ordered third-order self-energy diagrams with one pp rung (block II, time upward, dashed lines: bare interactions, external legs drawn horizontally, so that each horizontal cut between two interactions crosses only the lines of one intermediate state). (a) Rung at $t_{1'} < \tau < t_1$: the pair propagates in $\mathcal{P}$. (b) Rung at $\tau > t_1$: at t_1, the pair slot holds the holes k and l, the reversed configuration $(kl;i)$ in $\bar{\mathcal{P}}_{\rm pp}$, and the rung gives the first-order amplitude t^{ab}_{kl} of $U^{(2)}$.

The diagram in Fig. 3(b) and its mirror image, with the rung before $t_{1'}$, are the GF3 terms of Eqs. (D21) and (D22); the eh analogues are Eqs. (D27) and (D28). They arise from the time orderings of the three-particle propagator in which the pair lines run outside $[t_{1'}, t_1]$. When the pair interactions are resummed, reversed configurations appear in two places:

- At the two outer vertices, when the ladder or ring of a pair continues beyond t_1 or before $t_{1'}$. These configurations are folded into the couplings, $\mathcal{U} = U^{(1)} + \sum_\alpha \bar{U}^{(1)}_\alpha T_\alpha$, and enter Σ from third order on.
- Inside $[t_{1'}, t_1]$, within one pair, when the pair turns backward in time and forward again. These orderings give the terms $B_\alpha T_\alpha$ in $\mathcal{H}$ and the metric $\mathcal{N} = \mathbf{1} - \sum_\alpha T^\dagger_\alpha T_\alpha$, and enter Σ from fourth order on.

Between two different pairs, reversed configurations do not occur, since the no-overlap approximation separates consecutive pairs by a 2p1h cut, the factor $\mathcal{G}^{-1}$ in Eq. (36).

We call $\mathcal{P}$ the 2p1h and 2h1p configurations of Eq. (46), and $\bar{\mathcal{P}}_\alpha$ the reversed configurations of the pair α. Then:

- $\mathcal{G}$ lives on $\mathcal{P}$ only.
- Q_α lives on $\mathcal{P} \oplus \bar{\mathcal{P}}_\alpha$. We keep the symbol Q_α for its block on $\mathcal{P}$ and write $[Q_\alpha]_{\bar{\mathcal{P}}_\alpha \mathcal{P}}$ for its block from $\mathcal{P}$ to $\bar{\mathcal{P}}_\alpha$.
- The bare interaction couples the orbital p to $\mathcal{P}$ through $U^{(1)}$, Eq. (48), and to $\bar{\mathcal{P}}_\alpha$ through antisymmetrized integrals $\bar{U}^{(1)}_\alpha$.

Eq. (28) is therefore a sum over the blocks of R,

$$\Sigma^c(\omega) = \sum_{X,Y} V_X \left[R(\omega)\right]_{XY} V_Y^\dagger \ , \tag{50}$$

with $X, Y \in \{\mathcal{P}, \bar{\mathcal{P}}_{\rm pp}, \bar{\mathcal{P}}_{\rm eh}, \bar{\mathcal{P}}_{\overline{\rm eh}}\}$, $V_{\mathcal{P}} = U^{(1)}$ and $V_{\bar{\mathcal{P}}_\alpha} = \bar{U}^{(1)}_\alpha$. For $R = \mathcal{G}$, only $X = Y = \mathcal{P}$ contributes, and $\Sigma^c = U^{(1)} \mathcal{G}\, U^{(1)\dagger} = \Sigma^{(2)}$.

The block on $\mathcal{P}$ is $R_{\rm oc}$, Eq. (45). The other blocks follow in three steps.

First, in Eq. (36), consecutive factors are joined by $\mathcal{G}^{-1}$, which lives on $\mathcal{P}$. Only the first factor of a term can reach $\bar{\mathcal{P}}_\alpha$, and only if it is $Q_\alpha - \mathcal{G}$. These terms make up R_α, Eq. (38), so that

$$[R]_{\bar{\mathcal{P}}_\alpha\mathcal{P}} = [Q_\alpha]_{\bar{\mathcal{P}}_\alpha\mathcal{P}}\,\mathcal{G}^{-1}(R_{\rm oc} - R_\alpha)\;. \tag{51}$$

Second, we eliminate R_α. Adding $R_{\rm oc} - R_\alpha$ to both sides of Eq. (38) gives

$$R_{\rm oc} = Q_\alpha\,\mathcal{G}^{-1}(R_{\rm oc} - R_\alpha)\;,$$

i.e.

$$\mathcal{G}^{-1}(R_{\rm oc} - R_\alpha) = Q_\alpha^{-1} R_{\rm oc}\;.$$

Inserted into Eq. (51),

$$[R]_{\bar{\mathcal{P}}_\alpha\mathcal{P}} = [Q_\alpha]_{\bar{\mathcal{P}}_\alpha\mathcal{P}}\,Q_\alpha^{-1}\,R_{\rm oc}\;.$$

Third, the same argument for the last factor of a term gives the columns, and both together the corners,

$$\begin{aligned}
[R]_{\mathcal{P}\bar{\mathcal{P}}_\beta} &= R_{\rm oc}\,Q_\beta^{-1}[Q_\beta]_{\mathcal{P}\bar{\mathcal{P}}_\beta}\;,\\
[R]_{\bar{\mathcal{P}}_\alpha\bar{\mathcal{P}}_\beta} &= [Q_\alpha]_{\bar{\mathcal{P}}_\alpha\mathcal{P}}\,Q_\alpha^{-1}\,R_{\rm oc}\,Q_\beta^{-1}[Q_\beta]_{\mathcal{P}\bar{\mathcal{P}}_\beta}\;.
\end{aligned}$$

All blocks of R are thus $R_{\rm oc}$, multiplied from the left by $[Q_\alpha]_{\bar{\mathcal{P}}_\alpha\mathcal{P}}Q_\alpha^{-1}$ for a row in $\bar{\mathcal{P}}_\alpha$ and from the right by its adjoint for a column in $\bar{\mathcal{P}}_\beta$. Inserted into Eq. (50), the sum over the blocks factorizes,

$$\Sigma^c(\omega) = \mathcal{U}\,R_{\rm oc}(\omega)\,\mathcal{U}^\dagger\;,$$

with the couplings

$$\mathcal{U} = U^{(1)} + \sum_\alpha \bar{U}_\alpha^{(1)}[Q_\alpha]_{\bar{\mathcal{P}}_\alpha\mathcal{P}}\,Q_\alpha^{-1}\;. \tag{52}$$

The reversed configurations are thus folded into the couplings, and $R_{\rm oc}$ is needed only on $\mathcal{P}$. It remains to specify the three-particle propagators in the pp and eh channels, which we do in the next section.

B. Pair and three-particle propagators

The Faddeev series correlates two particles at a time, while the third one is a spectator. The three-particle propagators Q_α of Eq. (45) therefore follow from two-body pair problems and the free propagation of the spectator. The two-body problems are well known for the herein used first-order Faddeev kernel.[15] The eh propagator is obtained by solving the TDHF equations Eq. (B6), and the $\overline{\rm eh}$ propagator follows by crossing. Analogously, the pp propagator follows from the solution of the ppRPA, Eq. (B7). Coupled to two external orbitals, they give the two-body pair vertices $M^{\rm eh}_{pq,n}$ and $M^{\rm pp}_{pq,n}$ of the reducible vertices, Eqs. (B8) and (B9).

With one interacting pair, the three-particle propagator is the pair propagator times the free propagation of the spectator. For the pp pair with spectator r,

$$\begin{aligned}
Q_{{\rm pp};xyr,uwr}(\omega) = n_r \sum_{n\in N+2} \frac{Z^n_{xy}Z^n_{uw}}{\omega + \epsilon_r - \Omega^+_n + {\rm i}0^+}\\
+ \bar{n}_r \sum_{m\in N-2} \frac{Z^m_{xy}Z^m_{uw}}{\omega + \epsilon_r - \Omega^-_m - {\rm i}0^+}\;,
\end{aligned} \tag{53}$$

and for the eh pair with spectator s,

$$\begin{aligned}
Q_{{\rm eh};xrs,yus}(\omega) = \bar{n}_s \sum_{\Omega_n>0} \frac{Z^n_{xr}Z^n_{yu}}{\omega - \epsilon_s - \Omega_n + {\rm i}0^+}\\
+ n_s \sum_{\Omega_n>0} \frac{\tilde{Z}^n_{xr}\tilde{Z}^n_{yu}}{\omega - \epsilon_s + \Omega_n - {\rm i}0^+}\;.
\end{aligned} \tag{54}$$

Here, Z^n_{xy} are the components of the two-body eigenvectors of Eq. (B5) on the pair indices (xy), forward (X) or backward (Y). $Q_{\overline{\rm eh}}$ follows from $Q_{\rm eh}$ with the lines 1 and 2 exchanged. For a particle spectator only the forward eh excitations Z_n survive, for a hole spectator only the backward ones, $\tilde{Z}_n$.

Both propagators are block diagonal in 2p1h ⊕ 2h1p. The occupation factors select the first terms of Eqs. (53) and (54) in block II, where the spectator of the pp pair is the hole i and the spectators of the eh and $\overline{\rm eh}$ pairs are the particles b and a. The second term is selected in block I, where the spectator of the pp pair is the particle a and the spectators of the eh and $\overline{\rm eh}$ pairs are the holes j and i. Between the configurations of 2p1h ⊕ 2h1p, Eqs. (53) and (54) read, in matrix notation,

$$Q_{\rm pp}(\omega) = Z^{\rm II}_{\rm pp}(\omega + \epsilon_i - \mathbf{\Omega}^+)^{-1}Z^{{\rm II}\dagger}_{\rm pp} \;\oplus\; Z^{\rm I}_{\rm pp}(\omega + \epsilon_a - \mathbf{\Omega}^-)^{-1}Z^{{\rm I}\dagger}_{\rm pp}\;, \tag{55}$$

with the $N+2$ solutions in block II and the $N-2$ solutions in block I, and

$$\begin{aligned}
Q_{\rm eh}(\omega) &= Z_{\rm eh}(\omega - \epsilon_b - \mathbf{\Omega})^{-1}Z^\dagger_{\rm eh} \;\oplus\; Z_{\rm eh}(\omega - \epsilon_j + \mathbf{\Omega})^{-1}Z^\dagger_{\rm eh}\;,\\
Q_{\overline{\rm eh}}(\omega) &= Z_{\overline{\rm eh}}(\omega - \epsilon_a - \mathbf{\Omega})^{-1}Z^\dagger_{\overline{\rm eh}} \;\oplus\; Z_{\overline{\rm eh}}(\omega - \epsilon_i + \mathbf{\Omega})^{-1}Z^\dagger_{\overline{\rm eh}}\;.
\end{aligned} \tag{56}$$

Here, Z_α contains the components of the pair eigenvectors on the configurations of the block, $(ab;i)$, $(ia;b)$ in block II and $(ij;a)$, $(ia;j)$ in block I. The components on the remaining configurations of the pair, $\bar{Z}_\alpha$, enter only at the external vertices, Sec. III C.

1. *Amplitude Form of Propagators*

It is useful to replace the propagators above by a form that uses CC-amplitudes. This is possible as the RPA eigenvalue problem in each channel can equivalently be expressed as the Riccati equation (see Appendix C, and Refs. 9,16,17)

$$B^\dagger + DT + TA + TBT = 0\;. \tag{57}$$

The amplitudes map the forward components on the backward components,

$$Y = TX\;. \tag{58}$$

Eq. (57) is the ring-CCD (eh) and ladder-CCD (pp) amplitude equation of the channel.[9] The relations between GW, the RPA and (equation-of-motion) coupled cluster theory are analyzed in Refs. 16,18–20. Each channel introduces its own set of amplitudes, but they remain decoupled. Full CCD couples them through the crossed-ring and mosaic terms of the amplitude equation.[17] These couplings play the role of the crossed channels that the own-channel approximation neglects.

One can then show the relations

$$X\mathbf{\Omega}X^{-1} = A + BT \ . \tag{59}$$

and

$$XX^\dagger = (\mathbf{1} - T^\dagger T)^{-1} \ . \tag{60}$$

These relations are readily upgraded to the three-particle space in each channel,

$$\begin{aligned} \bar{Z}_\alpha &= T_\alpha Z_\alpha \ , \\ Z_\alpha \mathbf{\Omega}_\alpha Z_\alpha^{-1} &= A_\alpha + B_\alpha T_\alpha \ , \\ Z_\alpha Z_\alpha^\dagger &= (\mathbf{1} - T_\alpha^\dagger T_\alpha)^{-1} \ , \end{aligned} \tag{61}$$

with $\mathbf{\Omega}_{\text{pp}} = \mathbf{\Omega}^+$ and $\mathbf{\Omega}_{\text{eh}} = \mathbf{\Omega}_{\overline{\text{eh}}} = \mathbf{\Omega}$.

For block II (block I follows equivalently), Eqs. (55) and (56) are $Q_\alpha = Z_\alpha(\omega - s_\alpha - \mathbf{\Omega}_\alpha)^{-1} Z_\alpha^\dagger$, with the spectator energies

$$s_{\text{pp}} = -\epsilon_i \ , \qquad s_{\text{eh}} = \epsilon_b \ , \qquad s_{\overline{\text{eh}}} = \epsilon_a \ . \tag{62}$$

With Eq. (61), it can be written as

$$\begin{aligned} Q_\alpha(\omega) &= Z_\alpha(\omega - s_\alpha - \mathbf{\Omega}_\alpha)^{-1} Z_\alpha^{-1}\, Z_\alpha Z_\alpha^\dagger \\ &= [Z_\alpha(\omega - s_\alpha - \mathbf{\Omega}_\alpha) Z_\alpha^{-1}]^{-1} Z_\alpha Z_\alpha^\dagger \\ &= (\omega - s_\alpha - Z_\alpha \mathbf{\Omega}_\alpha Z_\alpha^{-1})^{-1} Z_\alpha Z_\alpha^\dagger \\ &= (\omega - s_\alpha - A_\alpha - B_\alpha T_\alpha)^{-1} (\mathbf{1} - T_\alpha^\dagger T_\alpha)^{-1} \ . \end{aligned} \tag{63}$$

Next, the spectator energy and the orbital energies in A_α add up to K, and the interaction part of A_α is the pair interaction C_α^{II}. For the pp pair in the configuration $(ab;i)$ and the eh pair in $(ia;b)$,

$$\begin{aligned} s_{\text{pp}} + A^{\text{pp}}_{ab,cd} &= (\epsilon_a + \epsilon_b - \epsilon_i)\delta_{ac}\delta_{bd} + \langle ab||cd\rangle \ , \\ s_{\text{eh}} + A^{\text{eh}}_{ia,jc} &= (\epsilon_a + \epsilon_b - \epsilon_i)\delta_{ac}\delta_{ij} + \langle aj||ic\rangle \ , \end{aligned}$$

i.e. $s_\alpha + A_\alpha = K + C_\alpha^{\text{II}}$, and

$$Q_\alpha(\omega) = (\omega - K - C_\alpha^{\text{II}} - B_\alpha T_\alpha)^{-1} (\mathbf{1} - T_\alpha^\dagger T_\alpha)^{-1} \ .$$

After a similar treatment for block I, and with

$$\begin{aligned} C_\alpha &= C_\alpha^{\text{II}} \oplus (-C_\alpha^{\text{I}}) \ , \\ B_\alpha T_\alpha &= B_\alpha^{\text{II}} T_\alpha^{\text{II}} \oplus (-B_\alpha^{\text{I}} T_\alpha^{\text{I}}) \ , \end{aligned} \tag{64}$$

both blocks combine into

$$Q_\alpha(\omega) = (\omega - K - C_\alpha - B_\alpha T_\alpha)^{-1} (\mathbf{1} - T_\alpha^\dagger T_\alpha)^{-1} \ . \tag{65}$$

In the Tamm–Dancoff limit, $B_\alpha = 0$ and $T_\alpha = 0$, and

$$Q_\alpha(\omega) = (\omega - K - C_\alpha)^{-1} \ , \tag{66}$$

where $K + C_\alpha$ is the Tamm–Dancoff Hamiltonian of the pair, shifted by the spectator energy.

C. External vertices and effective Hamiltonian

We are now in a position to state the central result of these derivations. Inserting the amplitude form of Q_α defined by Eq. (65) into Eq. (45), we obtain

$$\begin{aligned} R_{\text{oc}}^{-1}(\omega) =& \omega\, \mathcal{N} - \mathcal{H} \ , \\ \mathcal{H} =& K + C + \sum_\alpha B_\alpha T_\alpha \\ & - \sum_\alpha T_\alpha^\dagger T_\alpha (K + C_\alpha + B_\alpha T_\alpha) \ , \\ \mathcal{N} =& \mathbf{1} - \sum_\alpha T_\alpha^\dagger T_\alpha \ , \end{aligned} \tag{67}$$

in both blocks. $\mathcal{H}$ and the metric $\mathcal{N}$ are Hermitian and block diagonal,

$$\mathcal{N} = \mathcal{N}^{\text{II}} \oplus \mathcal{N}^{\text{I}} \ , \qquad \mathcal{H} = \mathcal{H}^{\text{II}} \oplus (-\mathcal{H}^{\text{I}}) \ .$$

For the couplings, Eq. (52), we need the blocks of the three-particle propagators between $\bar{\mathcal{P}}_\alpha$ and $\mathcal{P}$. With all rows of the pair eigenvectors, Eqs. (53) and (54) have the block $\bar{Z}_\alpha(\omega - s_\alpha - \mathbf{\Omega}_\alpha)^{-1} Z_\alpha^\dagger$ between $\bar{\mathcal{P}}_\alpha$ and $\mathcal{P}$. With $\bar{Z}_\alpha = T_\alpha Z_\alpha$, Eq. (61),

$$[Q_\alpha]_{\bar{\mathcal{P}}_\alpha \mathcal{P}} = T_\alpha Z_\alpha (\omega - s_\alpha - \mathbf{\Omega}_\alpha)^{-1} Z_\alpha^\dagger = T_\alpha Q_\alpha \ ,$$

so that Eq. (52) becomes $\mathcal{U} = U^{(1)} + \sum_\alpha \bar{U}_\alpha^{(1)} T_\alpha$. With all indices given explicitly, the couplings become

$$\begin{aligned} \mathcal{U} &= (\mathcal{U}^{\text{II}}, \mathcal{U}^{\text{I}}) = U^{(1)} + \sum_\alpha \bar{U}_\alpha^{(1)}\, T_\alpha \ , \\ \mathcal{U}^{\text{II}}_{p;abi} &= \langle pi||ab\rangle + \hat{P}^-_{ab} \sum_{cj} \langle pc||jb\rangle\, T^{\text{eh}}_{jc,ia} + \sum_{k<l} \langle pi||kl\rangle\, T^{\text{pp}}_{kl,ab} \ , \\ \mathcal{U}^{\text{I}}_{p;ija} &= \langle pa||ij\rangle - \hat{P}^-_{ij} \sum_{kb} \langle pk||bj\rangle\, T^{\text{eh}}_{ib,ka} + \sum_{b<c} \langle pa||bc\rangle\, T^{\text{pp}}_{ij,bc} \ . \end{aligned} \tag{68}$$

Here, $\bar{U}_\alpha^{(1)}$ couples to the backward configurations of the pair α (all but the first terms). In contrast to ADC(3), the couplings contain the channel-restricted CCD amplitudes instead of the first-order doubles ones. With the substitutions

$$T^{\text{eh}}_{jc,ia} \to t^{ac}_{ij} \tag{69}$$

and

$$T^{\text{pp}}_{kl,ab} \to t^{ab}_{kl} \ , \tag{70}$$

these are the ADC(3) couplings, that describe the first-order coupling of the backward amplitudes to single-particle states.

Projected onto $a < b$ and $i < j$, and with

$$\omega\mathcal{N} - \mathcal{H} = \mathcal{N}^{1/2}(\omega - H_{\text{eff}})\mathcal{N}^{1/2}$$

for positive definite $\mathcal{N}$, we obtain the self-energy

$$\begin{aligned} \Sigma^c(\omega) &= \mathcal{U}\, R_{\text{oc}}(\omega)\, \mathcal{U}^\dagger = \mathcal{U}(\omega\,\mathcal{N} - \mathcal{H})^{-1}\mathcal{U}^\dagger \\ &= \left(\mathcal{U}\,\mathcal{N}^{-1/2}\right)(\omega - H_{\text{eff}})^{-1}\left(\mathcal{U}\,\mathcal{N}^{-1/2}\right)^\dagger \ , \\ H_{\text{eff}} &= \mathcal{N}^{-1/2}\mathcal{H}\,\mathcal{N}^{-1/2} \ , \end{aligned} \tag{71}$$

which, written out block by block, is Eq. (29). The quasiparticle energies are the eigenvalues of the corresponding Hermitian Dyson supermatrix

$$\begin{pmatrix} h_0 + \Sigma(\infty) & \mathcal{U}\,\mathcal{N}^{-1/2} \\ \mathcal{N}^{-1/2}\,\mathcal{U}^\dagger & H_{\text{eff}} \end{pmatrix},$$

with h_0 the one-body Hamiltonian of Eq. (2). In block II, the matrix elements of Eq. (67) are

$$\begin{aligned} \mathcal{H}_{abi,cdj} &= K^{\text{II}}_{abi,cdj} + C^{\text{II}}_{abi,cdj} + \delta_{ij}\, h^{\text{pp}}_{ab,cd} \\ &\quad + \hat{P}^-_{ab}\hat{P}^-_{cd}\,\delta_{bd}\, h^{\text{eh}}_{ia,jc}\,, \\ \mathcal{N}_{abi,cdj} &= \delta_{ac}\delta_{bd}\delta_{ij} - \delta_{ij}\sum_{k<l} T^{\text{pp}}_{kl,ab} T^{\text{pp}}_{kl,cd} \\ &\quad - \hat{P}^-_{ab}\hat{P}^-_{cd}\,\delta_{bd}\sum_{ke} T^{\text{eh}}_{ke,ia} T^{\text{eh}}_{ke,jc}\,, \end{aligned} \tag{72}$$

with

$$\begin{aligned} h^{\text{pp}}_{ab,cd} &= \sum_{k<l}\Big[\,\langle ab\|kl\rangle\; T^{\text{pp}}_{kl,cd} + T^{\text{pp}}_{kl,ab}\;\langle kl\|cd\rangle\,\Big] \\ &\quad + \sum_{k<l}\sum_{m<n} T^{\text{pp}}_{kl,ab}(D^{\text{pp}}_{kl,mn} + \epsilon_i\,\delta_{km}\delta_{ln}) T^{\text{pp}}_{mn,cd}\,, \\ h^{\text{eh}}_{ia,jc} &= \sum_{ke}\Big[\,\langle ae\|ik\rangle\; T^{\text{eh}}_{ke,jc} + T^{\text{eh}}_{ke,ia}\;\langle ce\|jk\rangle\,\Big] \\ &\quad + \sum_{ke}\sum_{lf} T^{\text{eh}}_{ke,ia}(A^{\text{eh}}_{ke,lf} - \epsilon_b\,\delta_{kl}\delta_{ef}) T^{\text{eh}}_{lf,jc}\,. \end{aligned} \tag{73}$$

D. Phonon form

It is instructive to write the effective Hamiltonian in terms of the pair excitations (phonons). We do this for block II; block I follows in the same way with the block-I parts of Eqs. (55) and (56). By Eqs. (45) and (67),

$$\begin{aligned} \omega\mathcal{N} - \mathcal{H} &= \sum_\alpha Q_\alpha^{-1} - 2\mathcal{G}^{-1}\,, \\ Q_\alpha^{-1} &= Z_\alpha^{-\dagger}(\omega - s_\alpha - \mathbf{\Omega}_\alpha) Z_\alpha^{-1}\,. \end{aligned} \tag{74}$$

Here, $\mathbf{\Omega}_\alpha$ is the diagonal matrix of the pair energies, $\mathbf{\Omega}^+$ for the pp pair and $\mathbf{\Omega}$ for the eh and $\overline{\text{eh}}$ pairs, and s_α is the spectator energy, Eq. (62). Its ω-linear part gives $\mathcal{N}$, its constant part $\mathcal{H}$. With Eqs. (56) and (55), as well as the normalization in Eq. (65), the quantities appearing in the self-energy defined by Eq. (71) are

$$\begin{aligned} \mathcal{N} &= \mathbf{1} + \sum_\alpha \Big[(Z_\alpha Z_\alpha^\dagger)^{-1} - \mathbf{1}\Big]\,, \\ \mathcal{H} &= K^{\text{II}} + C^{\text{II}} + \sum_\alpha \Big[Z_\alpha^{-\dagger}(\mathbf{\Omega}_\alpha + s_\alpha) Z_\alpha^{-1} - K^{\text{II}} - C_\alpha\Big]\,, \\ \mathcal{U} &= U^{(1)} + \sum_\alpha \Big[M_\alpha Z_\alpha^{-1} - U^{(1)}\Big]\,, \end{aligned} \tag{75}$$

with the three-body couplings of the pairs $M_\alpha = U^{(1)} Z_\alpha + \bar{U}^{(1)}_\alpha \bar{Z}_\alpha$. Their entries are the two-body vertices of Eqs. (B8) and (B9), with the spectator as second orbital index,

$$\begin{aligned} M_{\text{pp},p;ni} &= M^{\text{pp}}_{pi,n}\,, \\ M_{\text{eh},p;nb} &= -M^{\text{eh}}_{bp,n}\,. \end{aligned}$$

IV. RELATIONS TO VARIOUS SCHEMES TO CONSTRUCT ELECTRONIC SELF-ENERGIES

A. Tamm–Dancoff limit and ADC(3)

In the Tamm–Dancoff limit, $B_\alpha = 0$, Eq. (57) gives $T_\alpha = 0$, and the three-particle propagators reduce to Eq. (66). Eq. (45) then gives

$$R_{\text{oc}}(\omega) = (\omega - K - C)^{-1} = (\omega - K^{\text{II}} - C^{\text{II}})^{-1} \oplus (\omega + K^{\text{I}} + C^{\text{I}})^{-1}\,, \tag{76}$$

with $C = C_{\text{pp}} + C_{\text{eh}} + C_{\overline{\text{eh}}}$, *i.e.* $\mathcal{N} = \mathbf{1}$ and $\mathcal{H} = K + C$. Since $T_\alpha = 0$, the couplings reduce to $\mathcal{U} = U^{(1)}$, and the self-energy is that of ADC(2)-X,[21] also known as the 2ph-TDA,[22–24] or multi-channel Dyson equation (MCDE).[25–27]

ADC(3) is not the strict Tamm–Dancoff limit. Instead, the amplitudes are restricted to second order,

$$\mathcal{U} = U^{(1)} + U^{(2)} + O(v^3)\,.$$

The TDA-limit [ADC(2)-X] and ADC(3) differ, in principle, only in the couplings. However, since ADC(3) is usually constructed to be third order complete, $\Sigma(\infty)$ is chosen to be the third-order static self-energy, while ADC(2)-X keeps it at the Hartree–Fock level. For molecules, ADC(2)-X is a very crude approximation anyway, with large errors for charged excitations, while offering no tangible numerical advantage over ADC(3). Changing the static part has a non-negligible effect, but its accuracy remains poor either way.[28]

Recent work has used CC-amplitudes in ADC(3). The CCD-ADC(3) from Ref. 29 uses exactly Eq. (68), but solves the full CCD equations to couple all channel amplitudes. However, the effective Hamiltonian is kept at the TDA level. Their EN-ADC(3) is the linearized, diagonal version of the ladder-CCD couplings.

Superficially, this scheme seems related to recent work by Coveney [30], since also Coveney[30] uses the full CCD amplitudes. However, his construction differs in key points. His Hamiltonian is constructed in the CC similarity transform structure. It does not contain any metric and is non-Hermitian due to the similarity transformation. It also contains a line dressing that ADC(4) contains only at 4th order.

B. Connections to (vertex-corrected) *GW*

The *GW* approximation and other schemes derived in the framework of Hedin's equations[31] are formulated in the eh channel only. It is therefore instructive to retain only the eh pair in Eq. (75). The quantities entering the effective Hamiltonian

then simplify to

$$\begin{aligned}
\mathcal{N}^{\mathrm{II}}_{iab,jcd} &= [(Z_{\mathrm{eh}}Z_{\mathrm{eh}}^{\dagger})^{-1}]_{ia,jc}\,\delta_{bd}\ , \\
\mathcal{H}^{\mathrm{II}}_{iab,jcd} &= [Z_{\mathrm{eh}}^{-\dagger}(\mathbf{\Omega}_{\mathrm{eh}}+\epsilon_b)Z_{\mathrm{eh}}^{-1}]_{ia,jc}\,\delta_{bd}\ , \\
\mathcal{U}^{\mathrm{II}}_{p;iab} &= \sum_n M^{\mathrm{II}}_{\mathrm{eh},p;nb}\,[Z_{\mathrm{eh}}^{-1}]_{n,ia}\ , \\
\mathcal{N}^{\mathrm{I}}_{ia\,j,kbl} &= [(Z_{\mathrm{eh}}Z_{\mathrm{eh}}^{\dagger})^{-1}]_{ia,kb}\,\delta_{jl}\ , \\
\mathcal{H}^{\mathrm{I}}_{ia\,j,kbl} &= [Z_{\mathrm{eh}}^{-\dagger}(\mathbf{\Omega}_{\mathrm{eh}}-\epsilon_j)Z_{\mathrm{eh}}^{-1}]_{ia,kb}\,\delta_{jl}\ , \\
\mathcal{U}^{\mathrm{I}}_{p;ia\,j} &= \sum_n M^{\mathrm{I}}_{\mathrm{eh},p;nj}\,[Z_{\mathrm{eh}}^{-1}]_{n,ia}\ ,
\end{aligned} \tag{77}$$

where we have specified the indices explicitly for later reference. In block II, the pair (ia) has the spectator particle b, in block I the spectator hole j, and the pair and its spectator are independent. With Eq. (29),

$$\begin{aligned}
\omega\,\mathcal{N}^{\mathrm{II}} - \mathcal{H}^{\mathrm{II}} &= Z_{\mathrm{eh}}^{-\dagger}(\omega-\epsilon_b-\mathbf{\Omega}_{\mathrm{eh}})Z_{\mathrm{eh}}^{-1}\ , \\
\omega\,\mathcal{N}^{\mathrm{I}} + \mathcal{H}^{\mathrm{I}} &= Z_{\mathrm{eh}}^{-\dagger}(\omega-\epsilon_j+\mathbf{\Omega}_{\mathrm{eh}})Z_{\mathrm{eh}}^{-1}\ ,
\end{aligned}$$

and the self-energy becomes

$$\Sigma_{\mathrm{eh},pq}(\omega) = \sum_{nb}\frac{M^{\mathrm{II}}_{\mathrm{eh},p;nb}\,M^{\mathrm{II}*}_{\mathrm{eh},q;nb}}{\omega-\epsilon_b-\Omega_n} + \sum_{nj}\frac{M^{\mathrm{I}}_{\mathrm{eh},p;nj}\,M^{\mathrm{I}*}_{\mathrm{eh},q;nj}}{\omega-\epsilon_j+\Omega_n}\ . \tag{78}$$

The eh-first-order reducible vertices have been defined in Eq. (B8), with $M_{\mathrm{eh},p;nb} = -M^{\mathrm{eh}}_{bp,n}$. The self-energy Eq. (78) is completely equivalent to the PSD completion of the Σ^{TDHF} self-energy[32] from Ref. 33, therein denoted as Σ^{PSDII}. The only difference is that they construct Eq. (B8) with a statically screened TDHF kernel. This also illustrates the interpretation of the PSD self-energy of Ref. 33 as a "single-channel parquet"-type approximation.

1. Double counting

It has been remarked in Ref. 33 that Σ^{PSDII} double counts the $\Sigma^{(2)}$ contribution. This raises the natural question of how this can be the case, since the Faddeev-ADC(3) construction performed here is double-counting-free by construction.

To see how the double counting of $\Sigma^{(2)}$ arises, we take $v \to 0$. Then $Z_{\mathrm{eh}} \to \mathbf{1}$, $\bar{Z}_{\mathrm{eh}} \to 0$, $\Omega_n \to \epsilon_a - \epsilon_i$, $\mathcal{N} \to \mathbf{1}$, $\mathcal{H}^{\mathrm{II}} \to \Delta_{abi}$, $\mathcal{H}^{\mathrm{I}} \to -\Delta_{ija}$, and the couplings become those of Eq. (49), $\mathcal{U}^{\mathrm{II}}_{p;iab} \to \langle pi\|ab\rangle$ and $\mathcal{U}^{\mathrm{I}}_{p;ia\,j} \to \langle pa\|ij\rangle$. Eq. (78) gives

$$\begin{aligned}
\Sigma_{\mathrm{eh},pq} \to \sum_{abi}\frac{\langle pi\|ab\rangle\,\langle ab\|qi\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b} + \sum_{ija}\frac{\langle pa\|ij\rangle\,\langle ij\|qa\rangle}{\omega+\epsilon_a-\epsilon_i-\epsilon_j} \\
= 2\,\Sigma^{(2)}_{pq}\ .
\end{aligned} \tag{79}$$

Every 2p1h and 2h1p configuration appears twice: first, as the pair (ia) with spectator b and then as the pair (ib) with spectator a. Of course, it would be possible to subtract this contribution

$$\Sigma^{c} = \Sigma_{\mathrm{eh}} - \Sigma^{(2)}\ , \tag{80}$$

as is done in the FLEX approximation[13,14,34,35] and in the several flavors of combinations of the GW + T-matrix approximations.[36,37] This would however defeat the purpose of the PSD-construction, as the resulting self-energy is then not a sum of squares anymore and hence not PSD.

2. GW

Constructing Eq. (B8) with the standard (direct electron–hole) RPA, and keeping only the direct vertex, gives the GW approximation,

$$\Sigma_{GW}(\omega) = M^{\mathrm{D}}_{\mathrm{eh}}[\omega - s_{\mathrm{eh}} - \mathbf{\Omega}_{\mathrm{eh}}]^{-1}M^{\mathrm{D}^{\dagger}}\ . \tag{81}$$

Written as an eigenvalue problem in the 2p1h and 2h1p spaces, this is the upfolded form of GW.[38,39] The functional form of the GW self-energy is retained independent of the linear-response problem that is solved to construct the $M^{\mathrm{D}}_{\mathrm{eh}}$ and to obtain $\mathbf{\Omega}_{\mathrm{eh}}$. Usually, the direct RPA is used, which is the TDHF form of the RPA without the direct electron–hole interaction. In Hedin's framework, the GW approximation naturally goes together with the RPA, since both arise from the lowest-order vertex, $\Gamma = 1$, when the non-interacting propagator is calculated at the Hartree level. In practice, the non-interacting Green's function is almost always chosen to be a (generalized) Kohn–Sham one.[40] This, together with the availability of TDDFT codes, has motivated the use of TDDFT to construct Eq. (81) in GW calculations for solids,[41–50] but also in molecules,[49,51–54] and it is to be noted that local vertices from the same kernels have also been placed in the self-energy.[42–45,48–51,53–65] More recently, the solid state community has used statically screened TDHF, i.e. the usual BSE. This approach is quite successful as it lowers the overestimated band gaps of quasiparticle self-consistent GW in solids.[66–69] For molecules, this approach significantly worsens ionization potentials[70] compared to standard qsGW. For molecules, TDHF screening has been tested,[71,72] and even polarizabilities from EOM-CCSD.[73] Combined with HF Green's functions, they somewhat improve over the RPA. However, they are unbalanced and miss the corresponding vertex in the self-energy.[32]

3. Σ^{TDHF}

This naturally motivates the choice of the Σ^{TDHF} self-energy that has been introduced by Maggio and Kresse [74] and investigated later by several authors.[70,72,75–78] A first-order vertex constructed to satisfy the Ward identity had been used before in self-consistent calculations for atoms and small molecules.[79,80] It operates in Hedin's framework. It solves the BSE for the eh channel with the complete first-order vertex (TDHF), and uses the same vertex in the self-energy. This self-energy is the natural choice if based on a HF non-interacting reference, since the TDHF vertex is the functional derivative of the HF self-energy with respect to G.

Hedin's formalism operates with a self-energy with two vertices, but one of them stays bare at first order. The resulting

self-energies cannot be written in a Hermitian (PSD) resolvent form, which has been the motivation to introduce the PSD construction from above[33] in the first place. Of course, the idea of the PSD completion of the self-energy is much older.[81] Either way, the Σ^{TDHF} self-energy can be obtained by replacing one of the two vertices of Eq. (78) by its direct part $M^{\mathrm{D}}_{\mathrm{eh}}$,

$$\Sigma_{GWT}(\omega) = M^{\mathrm{D}}_{\mathrm{eh}}[\omega - s_{\mathrm{eh}} - \mathbf{\Omega}_{\mathrm{eh}}]^{-1} M^{\dagger}_{\mathrm{eh}} \,. \tag{82}$$

This clearly looks like a hybrid between the GW approximation Eq. (81) and the PSDII self-energy Eq. (78). From the perspective of the present review, such a hybrid approximation seems to make very little sense. We however emphasize that, unlike the PSDII self-energy, it is derived rigorously in Hedin's framework. It of course comes with the downside that it stops being PSD. It also eliminates the contribution from the eh-triplet channel. The eh-triplet channel turns out to be absolutely devastating for numerical accuracy. In fact, in Ref. 33, the triplet channel has been eliminated completely, and the resulting self-energy, coined PSD1, was found to give excellent agreement with experiment. This is of course not a very satisfactory ad hoc approximation, albeit a physically well-motivated one.

A significant upside of the Σ^{TDHF} self-energy is that it is double-counting-free, by virtue of the fact that Hedin's equations only create a single diagram once. However, at third order in perturbation theory, it only contains half of the eh contribution. In the correlation channel language adopted here, it contains the contribution from C_{eh} (through $U^{\mathrm{D}}\mathcal{G}\,C_{\mathrm{eh}}\,\mathcal{G}\,U^{(1)\dagger}$), but not from $C_{\overline{\mathrm{eh}}}$ (which is also the reason why it is double-counting-free at second order).

In Hedin's scheme, these missing diagrams would be added to the self-energy through the second-order vertex. In addition to the missing half of the eh channel, the second-order vertex also adds the missing pp channel to the self-energy; the T matrix is fully contained in Hedin's equations.[82] It is a feature of Hedin's equations that they never add diagrammatically complete sets of diagrams at an iteration order. This can be understood from the structure of the equations and has for instance been pointed out by Schindlmayr and Godby [83]. Being diagrammatically complete at a certain order, a quality closely linked but not equivalent to being crossing symmetric, is a feature we consider an important property of any theory of the electronic self-energy.

4. $G3W2$-type vertex corrections

Within Hedin's equations, some of the simplest possible vertex corrections arise from a first-order (3-point) vertex of the form[78,84]

$$\Gamma(123) = \delta(12)\delta(13) + \mathrm{i}W(21)G(13)G(32) \,, \tag{83}$$

that, inserted into Hedin's expression for Σ, results in the family of the so-called $G3W2$-type self-energies,

$$\Sigma_{xc}(11') = \Sigma^{GW} - G(14)W(1'4)G(42)G(21')W(21) \tag{84}$$

that have been investigated first for solids[85–89] and later for molecules.[78,90] By splitting the screened Coulomb interaction into the bare interaction and its polarizable part $W = v + W^{\mathrm{c}}$, the $G3W2$ term splits into 4 contributions, schematically

$$GWGWG = GvGvG + GW^{\mathrm{c}}GvG + GvGW^{\mathrm{c}}G + GW^{\mathrm{c}}GW^{\mathrm{c}}G \,. \tag{85}$$

The sum of the first two terms is known as the SOSEX-approximation.[91,92] Bruneval and Förster [78] have coined the sum of the first three terms the 2SOSEX-approximation, pointing out that it constitutes a more systematic approximation to the full $G3W2$. We note that also a statically screened $G3W2$ approximation has been investigated both for solids[93] and molecules,[94] and more recently all these approximations to $G3W2$ have been used in a fully self-consistent scheme by Pokhilko *et al.* [95] and assessed for molecules[96] (notice that the authors of this method denote the static approximation to $G3W2$ as $G3W2$). Resumming the statically screened $G3W2$ vertex to all orders as an eh ladder gives the screened eh T matrix, recast as a vertex.[84] With W in the static limit, as used in practice by Romaniello, Bechstedt, and Reining [84], this is the vertex of Σ^{BSE} of Ref. 32, but in the self-energy only, while the eh propagator is kept at the level of the direct RPA.

Without any vertex corrections to the inner vertex that is used to generate W, the performance of all of these approximations is exceptionally poor. The performance of the statically screened $G3W2$ self-energy is satisfactory, but only because the magnitude of its effect in QP energies is very small.[94] However, when paired with vertex corrections for W, $G3W2$-type approximations perform well.[32,93]

TABLE II. Approximations to Eq. (71) built from the pairs of Eq. (36): exact at second order, eh and pp parts of ADC(3) at third order, and positive semidefinite self-energy (Faddeev: for positive definite $\mathcal{N}$). Faddeev, eh only: Eq. (45) with $\mathcal{Q}_{\mathrm{pp}} \to \mathcal{G}$.

	$\Sigma^{(2)}$	eh	pp	PSD	Ref.
Σ_{eh} (PSD)	$2\,\Sigma^{(2)}$	$C_{\mathrm{eh}} + C_{\overline{\mathrm{eh}}}$	–	yes	33
$\frac{1}{2}\Sigma_{\mathrm{eh}}$	exact	$\frac{1}{2}(C_{\mathrm{eh}} + C_{\overline{\mathrm{eh}}})$	–	yes	33
$\Sigma_{\mathrm{eh}} - \Sigma^{(2)}$	exact	$C_{\mathrm{eh}} + C_{\overline{\mathrm{eh}}}$	–	no	this work
Σ_{GWT}	exact	C_{eh}	–	no	32,74
Σ_{FLEX}	exact	$C_{\mathrm{eh}} + C_{\overline{\mathrm{eh}}}$	C_{pp}	no	13,14
Faddeev-ADC(3)	exact	$C_{\mathrm{eh}} + C_{\overline{\mathrm{eh}}}$	C_{pp}	yes	5–8

5. PSD-version of $G3W2$-type self-energies

a. PSD-2SOSEX. Like all vertex-corrected self-energies derived in Hedin's formalism, also the ones of $G3W2$-type cannot be written in resolvent form. Instead, their pole structures are rather complicated. Recently, the PSD-completion procedure of Stefanucci *et al.* [81] has been applied to derive a PSD-version of the GW + 2SOSEX self-energy,[125] and it is interesting to investigate how it fits into the effective Hamiltonian framework of Faddeev-ADC(3). Like GW, the GW + 2SOSEX self-energy is of the form of Eq. (78) and can be written in the

TABLE III. Treatment and combination of correlation channels in self-energy approximations: second-order exactness in the bare interaction and positive semidefiniteness (PSD). Second order of the PSD completions in finite systems: Ref. 33 and this work.

Class	Type	Approximation	$\Sigma^{(2)}$ exact	PSD	Ref.
pp pair	T matrix	G_0T_0@ppRPA (one-shot)	yes	yes	15,97
	T matrix	$G_{\mathrm{RS}}T_{\mathrm{RS}}$ (renormalized singles)	yes	yes	98
	T matrix	first order of the pp Hedin equations	yes	yes	99
	T matrix	Kadanoff–Baym T matrix (self-consistent)	yes	yes	100–102
	T matrix	pp and eh T matrix in the G1–G2 scheme (GKBA)	yes	not discussed	103–106
	T matrix	generalized T matrix (self-consistent, Gorkov)	yes	yes	107,108
pp pair, screened kernel	screened T matrix	screened ladder (W in the rungs)	–	in general no	109–111
	screened T matrix	pp Hedin vertex corrections (perturbative in T)	yes	not discussed	99
eh pair	GW	GW	no ($\Sigma_{\mathrm{D}}^{(2)}$ only)	yes	31
	eh T matrix	eh T matrix	yes	no	15
	Σ^{TDHF} + PSD	PSDI	no	yes	33
	Σ^{TDHF} + PSD	PSDII (Σ_{eh})	no	yes	33
	vertex in W only	GW with beyond-RPA screening	no ($\Sigma_{\mathrm{D}}^{(2)}$ only)	yes	46,66,67,69,71,73
channel sums	GW + T matrix	FLEX	yes	no	13,14,34,35
	GW + T matrix	$GW + T$	yes	no	36,37
	GW + T matrix	statically screened FLEX (Hedin form)	yes	no	112
	charge + spin ladders	ladder DΓA, TPSC (lattice models)	–	–	113–115
	GW + spin-flip ladder	GWT (magnons)	–	–	116,117
channels coupled	all channels	dynamically screened ladder ($g_{123} \to 0$)	yes	no guarantee	118,119
	all channels	parquet (static reducible vertices, TDA)	yes	no guarantee	35
	all channels	parquet approximation (lattice models)	yes	no guarantee	120,121
three-particle	Faddeev	real-time Faddeev (NEGF, bare v)	yes	not discussed	122,123
	Faddeev	Faddeev-ADC(3) (RPA pairs, metric $\mathcal{N}$)	yes	yes	this work
PSD completions	vertex-corrected GW	half-diagram squares (GW + vertex)	no (finite systems)	yes	81,89,124
	vertex-corrected GW	GW+2SOSEX-psd	no	yes	125
	vertex-corrected GW	ADC-$G3W2$	no	yes	126
non-PSD vertex	$G3W2$-type	SOSEX	yes	no	91
	$G3W2$-type	2SOSEX	yes	no	78
	$G3W2$-type	$G3W2$	yes	no	78,87,95,96
	$G3W2$-type	statically screened $G3W2$	yes	no	93,94
	$G3W2$-type	screened T matrix as vertex	yes	no	84
	$GW\Gamma$	$GW\Gamma$ (TDHF vertex, Σ^{TDHF})	yes	no	32,74,77,79,80
	$GW\Gamma$	$GW\Gamma$ (BSE vertex, Σ^{BSE})	yes	no	32,70
	$GW\Gamma$	$GW\Gamma$ (local TDDFT vertex)	no	no	44,48,59
extended CC	vertex-corrected GW	ECC exchange corrections	no ($\Sigma_{\mathrm{D}}^{(2)}$ only)	not discussed	16

form of Eq. (81),

$$\Sigma^{\mathrm{PSD},p}_{pq}(\omega) = \sum_{c,bi} \frac{(M^{\mathrm{D}}_{pc,bi} + \widetilde{M}^{\mathrm{D}}_{pc,bi})(M^{\mathrm{D}}_{qc,bi} + \widetilde{M}^{\mathrm{D}}_{qc,bi})^*}{\omega - \epsilon_c - \Delta_{bi}}. \quad (86)$$

where

$$\widetilde{M}^{\mathrm{D}}_{pc,n} = \sum_{aj} M^{\mathrm{D}}_{aj,n} \left[\frac{\langle pa|jc\rangle}{\epsilon_a - \epsilon_j - \Omega_n} + \frac{\langle pj|ac\rangle}{\epsilon_a - \epsilon_j + \Omega_n} \right]. \quad (87)$$

Clearly, in their first-order limit, $\Omega_n \to \epsilon_b - \epsilon_i$ and $M^{\mathrm{D}}_{aj,n} \to \langle ab|ij\rangle$, Eq. (87) becomes

$$\widetilde{M}^{\mathrm{D},(1)}_{pc,bi} = \sum_{aj} \langle ab|ji\rangle \left[\frac{\langle pa|jc\rangle}{\epsilon_a + \epsilon_i - \epsilon_b - \epsilon_j} + \frac{\langle pj|ac\rangle}{\epsilon_a + \epsilon_b - \epsilon_i - \epsilon_j} \right]. \quad (88)$$

The second term in this expression is a non-antisymmetrized second-order doubles amplitude, that is contained in ADC(3), and its renormalized version Eq. (87) is completely contained in Faddeev-ADC(3) as part of the ring-CCD amplitudes. The first term contains instead the transition-energy difference

$$\Delta_{aj} - \Delta_{bi} = (\epsilon_a - \epsilon_j) - (\epsilon_b - \epsilon_i). \quad (89)$$

This is the denominator for first-order mixing of the two forward particle–hole transitions (bi) and (aj), and it is part neither of the pp nor the eh-type couplings in Eq. (68).

To see how it fits into Faddeev-ADC(3), define the direct electron–hole scattering matrix between the $2p1h$ configurations $(c;bi)$ and $(c;aj)$ by

$$C^{\mathrm{eh,D},(1)}_{c,bi;\,c,aj} \equiv -\langle ab|ji\rangle\,. \quad (90)$$

where c is a fixed spectator. Their unperturbed energies are $E_{c,bi} = \epsilon_c + \Delta_{bi}$ and $E_{c,aj} = \epsilon_c + \Delta_{aj}$. They enter the Faddeev-ADC(3) self-energy, Eq. (71), via the expansion of the resol-

vent $(\omega\mathcal{N}-\mathcal{H})^{-1}$ via Eq. (67) through the contribution

$$\begin{aligned}\Sigma^{(3)}_{\mathrm{fwd,pq}}(\omega) &= \left[U^{(1)}g(\omega)C^{\mathrm{eh},(1)}g(\omega)U^{(1)\dagger}\right]_{pq}\\ &= \sum_{c,bi,aj}\frac{m_{p;c,bi}C^{\mathrm{eh},(1)}_{c,bi;\,c,aj}m^{*}_{q;c,aj}}{(\omega-E_{c,bi})(\omega-E_{c,aj})}\\ &= -\sum_{c,bi,aj}\frac{\langle pb|ci\rangle\,\langle ab|ji\rangle\,\langle qa|cj\rangle}{(\omega-E_{c,bi})(\omega-E_{c,aj})}.\end{aligned} \tag{91}$$

The denominator of this term can be expanded through a partial fraction decomposition as

$$\frac{1}{(\omega-E_{c,bi})(\omega-E_{c,aj})} = -\frac{1}{\Delta_{aj}-\Delta_{bi}}\left[\frac{1}{\omega-E_{c,bi}}-\frac{1}{\omega-E_{c,aj}}\right], \tag{92}$$

which shows that it contains the denominator of the first term in Eq. (88). The numerators are the same as the ones of the third-order terms of Eq. (86),

$$\Sigma^{(3),\mathrm{PSD},p}_{pq}(\omega) = \sum_{c,bi}\frac{\widetilde{M}^{\mathrm{D},(1)}_{pc,bi}M^{\mathrm{D},(0)*}_{qc,bi}+M^{\mathrm{D},(0)}_{pc,bi}\widetilde{M}^{\mathrm{D},(1)*}_{qc,bi}}{\omega-\epsilon_c-\Delta_{bi}}. \tag{93}$$

Of course, by expanding the resolvent, one can match the terms in the PSD-2SOSEX and Eq. (71) order by order.

Like its PSDII sibling, PSD-2SOSEX double counts part of $\Sigma^{(2)}$. The origin is the first term of Eq. (87). For the phonon n that reduces to the transition (aj) for $v\to 0$, its denominator is the RPA shift of that transition,

$$\epsilon_a-\epsilon_j-\Omega_n = O(v)\,,$$

and the term is of first order, not of second. In Eq. (88) it is the diagonal term $(aj)=(bi)$, where $\Delta_{aj}-\Delta_{bi}=0$. With the singlet phonons and spin-adapted vertices of Refs. 125,126, $M^{\mathrm{D}}_{aj,n}/(\epsilon_a-\epsilon_j-\Omega_n)\to -1/\sqrt{2}$, and the vertex of Eq. (86) becomes, to first order,

$$M^{\mathrm{D}}_{pc,aj}+\widetilde{M}^{\mathrm{D}}_{pc,aj}\to\frac{1}{\sqrt{2}}\Big(2\,\langle pa|cj\rangle-\langle pa|jc\rangle\Big)\,,$$

the singlet antisymmetrized vertex of PSDI. Summed over a, c and j,

$$\Sigma^{\mathrm{PSD}} = \tfrac{5}{4}\,\Sigma^{(2)}_{\mathrm{D}}+2\,\Sigma^{(2)}_{\mathrm{X}}+O(v^3)\,,$$

the same second order as PSDI. The cross term $2\,\Sigma^{(2)}_{\mathrm{X}}$ is the second-order exchange that GW+2SOSEX-aug already counts twice;[125] completing the square adds $\frac{1}{4}\Sigma^{(2)}_{\mathrm{D}}$.

In summary, double counting aside, the PSD-2SOSEX self-energy is a (severely) channel-restricted subset of the Faddeev-ADC(3) self-energy, but not through a one-to-one correspondence of individual pieces. As it does not assume the TDA, it contains terms that are not contained in ADC(3).

b. PSD-G3W2. Marie, Tölle, and Loos [126] applied the ADC construction to the expansion of the $G3W2$ self-energy in W, with the goal to complete it into a PSD expression. The configurations are those of Eq. (77), with direct-RPA phonons, $K=\epsilon_a+\Omega_\nu$ $(\epsilon_i-\Omega_\nu)$. Their PSD-$G3W2$ effective Hamiltonian goes beyond the Faddeev-ADC(3) configuration space and adds 3p2h configurations that would show up in ADC(4) and ADC(5). It also correlates the 5-particle block, terms that would otherwise only show up in ADC(5). Of course, since they solve the RPA problem, their scheme goes beyond ADC, but it is reasonable to assume that their self-energy is obtained in Faddeev-ADC(5).

Also PSD-$G3W2$ double counts a small fraction of the second-order contribution. It contains $\frac{17}{16}\Sigma^{(2)}_{\mathrm{D}}+\Sigma^{(2)}_{\mathrm{X}}$ in total. All of the recently introduced approaches[33,125,126] to make self-energies derived in Hedin's method PSD seem to share this shortcoming, and it is unclear how it can be overcome. It is also reasonable to assume that the PSD completions of higher-order self-energies derived in Hedin's equations would also double count contributions at higher orders. As already pointed out, the double counted contributions can of course be subtracted, but this would make the self-energies non-PSD. For this reason, the current PSD-completions do not seem to constitute a path toward systematically improvable self-energy approximations.

c. Why PSD completions overcount. For real ω below all poles of a Hermitian self-energy, the resolvent

$$R(\omega) = (\omega-M)^{-1} \tag{94}$$

is negative definite (above all poles, positive definite), so that the self-energy is a Gram form,

$$\Sigma_{pp}(\omega) = \langle U_p, R\,U_p\rangle. \tag{95}$$

A PSD completion represents a term T of a non-PSD target self-energy as a cross term, by splitting a coupling into two parts,

$$U_p = a+b, \tag{96}$$

such that

$$T = 2\,\mathrm{Re}\,\langle a, R\,b\rangle. \tag{97}$$

The Hermitian form then delivers the two squares as well, and by the Cauchy–Schwarz inequality

$$\big|\langle a,R\,a\rangle+\langle b,R\,b\rangle\big| \geq |T|. \tag{98}$$

The square $\langle a,R\,a\rangle$ is usually a term of the target. The square $\langle b,R\,b\rangle$ is not. It cannot vanish, since T needs $b\neq 0$, and it cannot cancel against the cross terms, since it is itself of one sign. Completing the square to represent $2ab$ always adds b^2. This is harmless only if b^2 is of higher order than the truncation. In ADC it is: the partners of the cross terms are $U^{(2)}$, whose squares are of fourth order, while the squares within the truncation, $U^{(1)\dagger}g\,U^{(1)}$ including its first-order energy shift, are target terms with weight one. In the completions of self-energies derived from Hedin's equations, b is of lower order than its explicit power of v, because it contains a resummed propagator or a small denominator. In PSD-2SOSEX, $\widetilde{M}$ carries $\epsilon_a-\epsilon_j-\Omega_n=O(v)$ and is effectively of zeroth order, so that b^2 appears already at second order: b is the exchange part of the first-order vertex, and its square is the surplus $\frac{1}{4}\Sigma^{(2)}_{\mathrm{D}}$ found above. Subtracting the surplus destroys positive semidefiniteness. Absorbing it instead requires rematching every square that falls within the truncation, order by order, which is the ADC construction itself.

C. The pp channel

Very similar to working in the eh channel only, it is of course also possible to restrict Eq. (75) to the pp channel only. The canonical self-energy approximation that operates in this channel is the T-matrix approximation.[15,99] Unlike in the eh channel, there is no double counting ambiguity in the pp channel, simply due to the fact that it is not related by a simple symmetry to another channel in the same way as the eh channel. In the same way as the PSDII self-energy, the resulting self-energy is complete to third order in the pp channel, and of course, by construction, it is also PSD.

T-matrix self-energies have been used and extended in several communities. In molecules, the G_0T_0 approximation is built on the ppRPA,[97] on a renormalized-singles reference,[98] compared to GW and the eh T matrix,[15] used in the Bethe–Salpeter equation,[127] and obtained as the first order of the pp Hedin equations.[99] Self-consistent T matrices are used in Kadanoff–Baym dynamics,[100–102,104,105] and, with anomalous propagators, in nuclear matter.[107,108] Screening enters the pp kernel in the screened ladder approximation of plasma physics,[109–111] through the Gorkov–Melik-Barkhudarov correction,[128] and through the vertex corrections of the pp Hedin equations.[99]

D. FLEX and other combinations of eh and pp channels

Truncating the Faddeev series, Eq. (36), after the terms with a single interacting pair, Eq. (32), gives the one-shot FLEX approximation,[13,14]

$$R_{\text{FLEX}} = \mathcal{G} + \sum_\alpha (Q_\alpha - \mathcal{G}) = Q_{\text{pp}} + Q_{\text{eh}} + Q_{\overline{\text{eh}}} - 2\,\mathcal{G}\,.$$

Each pair couples to the one-particle states through its own couplings, $\mathcal{U}_\alpha = M_\alpha Z_\alpha^{-1}$, and the free propagation through $U^{(1)}$,

$$\Sigma_{\text{FLEX}} = \sum_\alpha \mathcal{U}_\alpha\, Q_\alpha\, \mathcal{U}_\alpha^\dagger - 2\,U^{(1)}\mathcal{G}\,U^{(1)\dagger}\,. \tag{99}$$

Each pair term contains $\Sigma^{(2)}$ once, and $-2\,\Sigma^{(2)}$ removes the excess. Faddeev-ADC(3) and FLEX agree through first order in C; both are exact through third order. FLEX however misses the sequences of different pairs in Eq. (36), which enter at fourth order. Since the subtraction is outside the resolvent, Σ_{FLEX} has negative residues and no Hermitian effective Hamiltonian. In the Tamm–Dancoff limit,

$$R_{\text{FLEX}} = \sum_\alpha (\omega - K - C_\alpha)^{-1} - 2(\omega - K)^{-1}\,.$$

FLEX is usually motivated as a drastic approximation to the parquet approximation,[14,35] but closely related approximations can also be motivated as a rather heuristic combination of correlation channels. For instance, to place eh and pp on the same footing, GW and the T matrix have been summed with a subtraction of the common second-order terms in $GW + T$.[36,37] Starting from Hedin's equations, also Cunningham[112] has rederived FLEX, but he suggested screening all pair channels statically in the TDA. In the eh channel he suggested to use the kernel $v - W$, and in the pp channel he screened all interactions ($W - W$, schematically). The same has been done for instance also by Romaniello and Berger[129]. Instead of being summed, the channels are coupled within one two-particle correlation function in the dynamically screened ladder approximation[118,119] and in the parquet approximation.[35,120,121]

The static screening approximation is of course motivated by the usual GW-BSE approximation for neutral excitations, and many authors have solved the eh channel RPA with a statically screened HF problem.[33,67,69,70,72,78] This is well motivated for solids, where TDHF is problematic and will even diverge for metals. For molecules, this replacement is motivated by the superior performance of GW-BSE over TDHF. However, due to the fact that static screening effects are usually rather weak in molecules, the effects on ionization potentials are minor.[28,32]

V. FADDEEV-ADC WITH THE SECOND-ORDER PARQUET KERNEL

Faddeev-ADC(3) resums, in every pair, the own-channel ladder of the first-order vertex $\Gamma^{(1)}$, Eq. (25). It is exact through third order in v. In this section, we include the second-order parquet kernel in the pairs and repeat the steps of Sec. III. The main difference is that the second-order Faddeev kernel is frequency-dependent, which affects the expressions of the three-particle propagators. The frequency-dependence opens five-line cuts, and the resulting effective Hamiltonian of Faddeev-ADC(4) must be extended by 3p2h/3h2p manifolds, in the same way as ADC(4). We will also see that recombining the three pairs as in Faddeev-ADC(3) leads to an effective Hamiltonian with a non-positive metric, which leads to spurious poles. We then show that, upon addition of certain terms, a Hermitian effective Hamiltonian with a positive metric is again obtained. The cost of the method is very similar to the one of ADC(4).

A. Second-order pair vertices

Inserting Eqs. (A4) and (A8) into Eq. (A3), we obtain

$$\begin{aligned}\Gamma_{\text{pp}} &= \Gamma^{(1)} + \Phi^{\text{eh},(2)} + \Phi^{\overline{\text{eh}},(2)}\,,\\ \Gamma_{\text{eh}} &= \Gamma^{(1)} + \Phi^{\text{pp},(2)} + \Phi^{\overline{\text{eh}},(2)}\,,\end{aligned} \tag{100}$$

and $\Gamma_{\overline{\text{eh}}}$ follows from Γ_{eh} by crossing. Each pair thus receives the second-order vertices of the two other channels. The own-channel BSEs, Eq. (25), become

$$\begin{aligned}F_{\text{oc}}^{\text{pp}} &= \Gamma_{\text{pp}} - \frac{1}{2}\,\Gamma_{\text{pp}} \circ_{\text{pp}} F_{\text{oc}}^{\text{pp}}\,,\\ F_{\text{oc}}^{\text{eh}} &= \Gamma_{\text{eh}} + \Gamma_{\text{eh}} \circ_{\text{eh}} F_{\text{oc}}^{\text{eh}}\,,\end{aligned} \tag{101}$$

which define, through Eq. (26), the series Eq. (27) with a second-order kernel. The second-order vertices of Eq. (100)

are the functional derivative of the second-order self-energy,[35] and we use them together with lines that contain $\Sigma^{(2)}$ in $\mathcal{G}$, Eq. (15). Fig. 4 shows both equations.

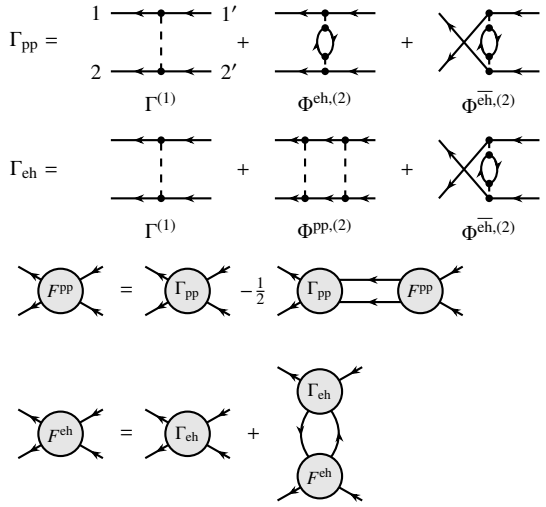


FIG. 4. The second-order pair vertices, Eq. (100), and the own-channel BSEs, Eq. (101). In the pp BSE, consecutive vertices are joined by two lines in the same direction, in the eh BSE by an eh pair.

In contrast to $\Gamma^{(1)}$, the vertices in Eq. (100) are frequency dependent. As shown in Figure 4, $\Phi^{\mathrm{eh},(2)}$ contains one eh bubble between two bare interactions, and $\Phi^{\mathrm{pp},(2)}$ one pp or hh pair. When both interactions lie between $t_{1'}$ and t_1, the horizontal cut between them cuts through five lines in a self-energy diagram. In block II, these are 3p2h configurations $J = (cde;kl)$, $c < d < e$, $k < l$, with the free energies

$$K_{3,J} = \epsilon_c + \epsilon_d + \epsilon_e - \epsilon_k - \epsilon_l \ . \tag{102}$$

Block I has the corresponding 3h2p configurations.

A five-line configuration between two interactions is not a 2p1h cut. It therefore belongs to the 2p1h-irreducible kernel Ξ of the Dyson equation $R^{-1} = \mathcal{G}^{-1} - \Xi$, Eq. (14), and not to the pair propagators Q_α of Eq. (65), which stay as they are. As a consequence, R^{-1} is no longer linear in ω, as in Eq. (67), but has poles at the five-line energies. In the same way as the R_{oc} of Eq. (67) can be written as the inverse of an effective 3-particle Hamiltonian, R can be written as the inverse of an effective, dynamical 3-particle Hamiltonian

$$R^{-1} = A - WD^{-1}W^\dagger \ , \tag{103}$$

that is the fold of the effective, static 5-particle Hamiltonian,

$$\begin{pmatrix} A & W \\ W^\dagger & D \end{pmatrix} = \omega\mathbf{N} - \mathbf{H} \ . \tag{104}$$

Here, $A = \omega\mathcal{N} - \mathcal{H} - \Delta C$ is the Faddeev-ADC(3) result, Eq. (67), together with the part ΔC of the line dressing without a five-line pole. It is obtained as

$$\Delta C^{\mathrm{II}}_{abi,cdj} = \delta_{ij}\,\hat{P}^-_{ab}\hat{P}^-_{cd}\,\delta_{bd}\,\sigma^<_{ac} - \delta_{ac}\delta_{bd}\,\sigma^>_{ij} \ , \tag{105}$$

where

$$\sigma^{\lessgtr}_{pq} = \frac{1}{2}[\Sigma^{\lessgtr}_{pq}(\epsilon_p) + \Sigma^{\lessgtr}_{pq}(\epsilon_q)] \tag{106}$$

Is the quasiparticelized GF2 self-energy.

$$\Sigma^<_{pq}(z) = \frac{1}{2}\sum_{klc}\frac{\langle pc||kl\rangle\,\langle kl||qc\rangle}{z + \epsilon_c - \epsilon_k - \epsilon_l} \ ,$$
$$\Sigma^>_{pq}(z) = \frac{1}{2}\sum_{kcd}\frac{\langle pk||cd\rangle\,\langle cd||qk\rangle}{z + \epsilon_k - \epsilon_c - \epsilon_d} \ .$$

the analogous expression is valid for block I.

It remains to specify the structure of the 5-particle block and its couplings to the three-particle states. D is the diagonal matrix of the 5-line states,

$$D = \omega - K_3 \ , \qquad (K_3)_{JJ'} = \delta_{JJ'}\,K_{3,J} \ . \tag{107}$$

The couplings specify the possible ways how a 3-line state can be turned into a 5-line state. This is only possible if a line emits a particle-hole pair by action of a Coulomb interaction. Denote a 3-line state by P, and a 5-line state by J, we can write the couplings as

$$W_{PJ} = \langle P|\hat{V}|J\rangle \ . \tag{108}$$

Their numerical values can be worked out by the Slater condon rules. With $|P\rangle = \hat{a}^\dagger_a\hat{a}^\dagger_b\hat{a}_i\,|\Phi_0\rangle$, $a < b$, and $|J\rangle = \hat{a}^\dagger_c\hat{a}^\dagger_d\hat{a}^\dagger_e\hat{a}_l\hat{a}_k\,|\Phi_0\rangle$, $c < d < e$, $k < l$, the Slater–Condon rules give

$$\langle P|\hat{V}|J\rangle = \tfrac{1}{2}\mathcal{P}^-_{cde}\hat{P}^-_{kl}\Big[\underbrace{\delta_{be}\delta_{il}\,\langle ak||cd\rangle}_{e_a} - \underbrace{\delta_{ae}\delta_{il}\,\langle bk||cd\rangle}_{e_b} + \underbrace{\delta_{ad}\delta_{be}\,\langle kl||ci\rangle}_{e_i}\Big] \ , \tag{109}$$

where

$$\hat{P}^-_{pq} = 1 - \hat{P}_{pq} \ ,$$
$$\mathcal{P}^-_{pqr} = (1 - \hat{P}_{pr} - \hat{P}_{qr})\hat{P}^-_{pq} \ , \tag{110}$$

are the usual anti-permutation operators. In e_u, the line u emits an eh pair and the two other lines of P reappear in J.

B. Positive-norm effective Hamiltonian

(block I with the line energies reversed). With Eq. (104),

$$\Sigma^{c,\mathrm{II}}(\omega) = \mathbf{U}(\omega\mathbf{N} - \mathbf{H})^{-1}\mathbf{U}^\dagger \ , \tag{111}$$

with

$$\mathbf{N} = \begin{pmatrix} \mathcal{N} & 0 \\ 0 & \mathbf{1} \end{pmatrix}, \qquad \mathbf{H} = \begin{pmatrix} \mathcal{H} + \Delta C & -W \\ -W^\dagger & K_3 \end{pmatrix},$$
$$\mathbf{U} = (\mathcal{U} + \Delta U,\ U_3) \ , \tag{112}$$

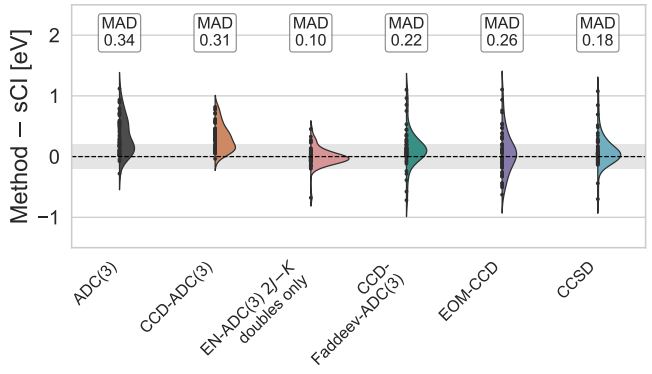


FIG. 5. Errors of the IPs with respect to sCI (aug-cc-pVTZ, frozen core) for ADC(3), CCD-ADC(3), EN-ADC(3), CCD-Faddeev-ADC(3) and CCSD, over all 58 states with an sCI value (second block of Table IV).

U_3 and ΔU are the remaining couplings of ADC(4).[11] $\mathbf{N} > 0$ whenever $\mathcal{N} > 0$, and

$$\Sigma^{c,\mathrm{II}}(\omega) = (\mathbf{U}\mathbf{N}^{-1/2})(\omega - \mathbf{H}_{\mathrm{eff}})^{-1}(\mathbf{U}\mathbf{N}^{-1/2})^{\dagger} ,$$
$$\mathbf{H}_{\mathrm{eff}} = \mathbf{N}^{-1/2}\mathbf{H}\,\mathbf{N}^{-1/2} = \begin{pmatrix} H_{\mathrm{eff}} + \mathcal{N}^{-1/2}\Delta C\,\mathcal{N}^{-1/2} & -\mathcal{N}^{-1/2}W \\ -W^{\dagger}\mathcal{N}^{-1/2} & K_3 \end{pmatrix}, \quad (113)$$

with H_{eff} of Eq. (71). We call this Faddeev-ADC(4).

Faddeev-ADC(4) lifts the no-overlap approximation at the five-line cuts. The Faddeev recombination of the three pairs, Eq. (45), with the second-order kernel and $\Sigma^{(2)}$ in the lines, contains the line dressing and the crossed vertices of Eq. (100), but not the time orderings in which the rungs of two different pairs overlap across a five-line cut. Without them, its residues at the 3p2h energies have negative eigenvalues (in a model with random antisymmetrized integrals, down to −0.54 times the largest), and, as for FLEX, Eq. (99), it has no Hermitian effective Hamiltonian.

VI. NUMERICAL RESULTS

Most of these approximations have never been evaluated for the same set of molecules in a reasonable basis set. Here, we therefore evaluate many of them for the set of 58 IPs of 23 small molecules for which Marie and Loos [130] have recently calculated reference values of near FCI quality in the aug-cc-pVQZ basis. We do not consider statically screened interactions to calculate the ph/pp RPA phonons. For once, these are heuristic adjustments, and, if not applied carefully, they can lead to double counting issues. Moreover, they can only be expected to have a minor influence on the final results. A method that does not perform well with bare interactions will not perform well with statically screened interaction either.

Table IV summarizes the errors, and all raw numerical data is shown in Table VI. All calculations use RHF references, are performed in an all-electron framework, make use of density fitting with the aug-cc-pVQZ-RI basis, and, except *GW*, add the MP2-relaxed static self-energy. Faddeev-ADC(3) and the ADC(3) variants are solved as Dyson supermatrices; FLEX, PSD and GF2 in the diagonal quasiparticle approximation. T1 and T0 denote the triplet eh channel at first order and switched off, respectively. C_2 and BN are excluded, since their RHF references are unstable in the TDHF singlet (and triplet) channel, and F_2, since its TDHF triplet is unstable. The statistics run over the 48 states with a value for every method. FLEX finds no quasiparticle root for NH_3 (3) (RPA) and Ne (2) (T0).

From a formal perspective, Faddeev-ADC(3) has many desirable properties. It is exact to third order, double-counting-free,

TABLE IV. Errors (eV) with respect to sCI over the 52 states of Table VI with a value for every method, and MAD over their 20 first IPs; aug-cc-pVTZ, frozen core. Static self-energy: MP2r, MP2 density, relaxed; MP2u, unrelaxed; MP3u, MP3 unrelaxed; /CCD and /EN, the MP2 functional evaluated on the density of the CCD or the EN-dressed amplitudes; –, none.

	N	MAD	MSE	max	MAD first IP	static
aug-cc-pVTZ, frozen core, 52 states with a value for every method (20 first IPs)[a]						
ADC(3)	52	0.34	+0.34	1.12	0.25	MP2r
CCD-ADC(3)	52	0.31	+0.31	0.82	0.23	MP2r/CCD
EN-ADC(3)	52	0.09	+0.01	0.45	0.05	MP2r/EN
Faddeev-ADC(3) RPA	52	1.94	+1.91	21.41	1.51	MP2r
Faddeev-ADC(3) T1	52	0.33	+0.31	0.71	0.29	MP2r
Faddeev-ADC(3) T0	52	0.61	+0.57	3.48	0.29	MP2r
CCD-Faddeev-ADC(3)	52	0.18	+0.11	1.10	0.11	MP2r/CCD
Faddeev-ADC(3) dRPA	52	0.80	+0.74	3.87	0.42	MP2r
FLEX RPA	52	1.36	−0.45	5.87	1.22	MP2r
FLEX T1	52	1.10	−1.02	3.72	0.85	MP2r
FLEX T0	52	0.60	+0.57	3.39	0.29	MP2r
FLEX CCD	52	1.21	−1.13	3.85	0.97	MP2r
PSD1	52	0.54	−0.07	3.26	0.32	MP2r
PSD1 $-D/4-X$	52	0.29	−0.04	1.73	0.19	MP2r
PSD2	52	2.37	−1.51	8.19	1.87	MP2r
PSD2 −GF2	52	1.65	−0.79	5.35	1.38	MP2r
PSD2 T1 −GF2	52	1.43	−1.35	4.22	1.15	MP2r
CCSD	52	0.18	+0.10	1.08	0.09	–
EOM-CCD	52	0.25	+0.09	1.10	0.19	–
GW	52	0.45	+0.42	2.07	0.34	–
GF2	52	0.79	−0.63	1.92	0.76	MP2r
aug-cc-pVTZ, frozen core, all 58 states with an sCI value (23 first IPs)						
ADC(3)	58	0.34	+0.33	1.12	0.28	MP2r
CCD-ADC(3)	58	0.31	+0.31	0.82	0.24	MP2r/CCD
EN-ADC(3)	58	0.10	−0.01	0.68	0.06	MP2r/EN
CCD-Faddeev-ADC(3)	58	0.22	+0.14	1.10	0.18	MP2r/CCD
Faddeev-ADC(3) dRPA	58	0.80	+0.68	3.87	0.47	MP2r
FLEX CCD	58	1.33	−1.10	4.53	1.15	MP2r
CCSD	58	0.18	+0.10	1.08	0.10	–
EOM-CCD	58	0.26	+0.07	1.10	0.22	–
GW	58	0.46	+0.40	2.07	0.35	–
GF2	58	0.83	−0.69	2.26	0.78	MP2r

[a] Errors with respect to sCI/aug-cc-pVTZ; frozen core as for the sCI references[130] (He core of B–Ne, Ne core of Na–Ar, Li and Be all-electron), DF aug-cc-pVTZ-RI, MP2-relaxed static self-energy (for EN-ADC(3), CCD-ADC(3) and CCD-Faddeev-ADC(3) from their own EN-dressed and CCD-amplitude densities, as in Ref. 29; FLEX with the closed-form pp term). C_2, BN and F_2 are excluded from the first block (unstable TDHF channel). Faddeev-ADC(3) RPA and FLEX RPA for BF, BeO, CS and N_2: roots in energy order, not assigned states. FLEX, PSD, GF2 and GW: the root of largest pole strength Z among all roots of the self-energy equation within 8 eV of the ADC(3) value of the same orbital; roots with $Z < 0.3$ are marked in Table VI.

TABLE V. Errors (eV) with respect to sCI/aug-cc-pVTZ, frozen core: Faddeev-ADC(4) and ADC(4) next to the third-order-complete methods, and all third-order-complete methods with the MP2-unrelaxed static self-energy. Static self-energy as in Table IV.

	N	MAD	MSE	max	MAD first IP	static
aug-cc-pVTZ, frozen core, 58 states with a value for every method (23 first IPs)[b]						
Faddeev-ADC(4) CCD	58	0.31	−0.23	2.42	0.31	MP3u
ADC(4)	58	0.35	−0.29	2.73	0.32	MP3u
ADC(3)	58	0.37	+0.34	1.77	0.28	MP2u
CCD-Faddeev-ADC(3)	58	0.24	+0.12	1.18	0.21	MP2u/CCD
The same, without BeO and LiF (54 states, 21 first IPs)						
Faddeev-ADC(4) CCD	54	0.22	−0.12	0.81	0.21	MP3u
ADC(4)	54	0.25	−0.18	0.83	0.20	MP3u
ADC(3)	54	0.31	+0.28	1.77	0.21	MP2u
CCD-Faddeev-ADC(3)	54	0.21	+0.08	1.18	0.18	MP2u/CCD
aug-cc-pVTZ, frozen core, MP2-unrelaxed static self-energy, every state each met						
ADC(3)	58	0.37	+0.34	1.77	0.28	MP2u
CCD-ADC(3)	58	0.30	+0.29	1.20	0.22	MP2u/CCD
EN-ADC(3)	58	0.12	−0.04	0.67	0.09	MP2u/EN
Faddeev-ADC(3) RPA	52	1.93	+1.91	21.04	1.54	MP2u
Faddeev-ADC(3) T1	55	0.35	+0.32	1.53	0.28	MP2u
Faddeev-ADC(3) T0	55	0.65	+0.56	3.43	0.34	MP2u
CCD-Faddeev-ADC(3)	58	0.24	+0.12	1.18	0.21	MP2u/CCD
Faddeev-ADC(3) dRPA	58	0.81	+0.68	3.82	0.48	MP2u

PSD, Hermitian, and includes renormalized amplitudes via the solution of the ring- and ladder-CCD equations. However, its numerical performance is abysmal, coming with a MAD of 2.05 eV. FLEX performs nearly equally badly, and the only method that performs even worse than Faddeev-ADC(3) is the PSDII approximation of Ref. 33. What is the reason for this bad performance of these methods? All of them have in common that they include TDHF triplet amplitudes. The negative effect on performance has already been observed by Pavlyukh, Bruneval, and Förster[33], who then introduced the PSD1 approximation, that omits triplet excitations altogether. This improves the MAD from more than 2.0 eV to about 0.5 eV, which is not great, but somewhat reasonable, in the same ballpark as G_0W_0. By making it double-counting-free, its MAD nearly halves, to 0.29 eV, on par with ADC(3).

This motivates evaluating Faddeev-ADC(3) and FLEX without triplet excitations. When the triplet channel is omitted altogether (T0), both methods significantly improve, and the MADs become 0.51 and 0.49 eV. As a less drastic approximation, one can replace the triplet eh amplitudes by their first-order doubles counterparts that are also used in ADC(3). For Faddeev-ADC(3), this reduces the error further to 0.33 eV. For FLEX it does not help; the error increases again to more than 1.0 eV. Another way to avoid triplet instabilities is to use direct RPA instead of TDHF in the eh channel. This can be done both for Faddeev-ADC(3) and FLEX, but we have only numerically evaluated it for Faddeev-ADC(3). For the subset of 48 states for which Faddeev-ADC(3) is defined this gives a MAD of 0.67 eV, which is somewhat more reasonable than the 2.05 eV of Faddeev-ADC(3) with the RPA channels, but still far from good.

The results clearly show that channel coupling is important. The same has been found for correlation energies in coupled-cluster theory, where each single channel of CCD, rings, ladders or crossed rings, individually, diverges for the ~~electron~~ gas. Only the terms that couple the channels (the mosaic terms) remove the divergence.[9,17] The reason is that ~~the trip~~let eh channel resummed with the bare interaction is close to unstable, and sometimes even completely unstable, as for instance seen for F_2. As shown in Fig. 6, the error of Faddeev-ADC(3) grows as the lowest TDHF triplet approaches zero.

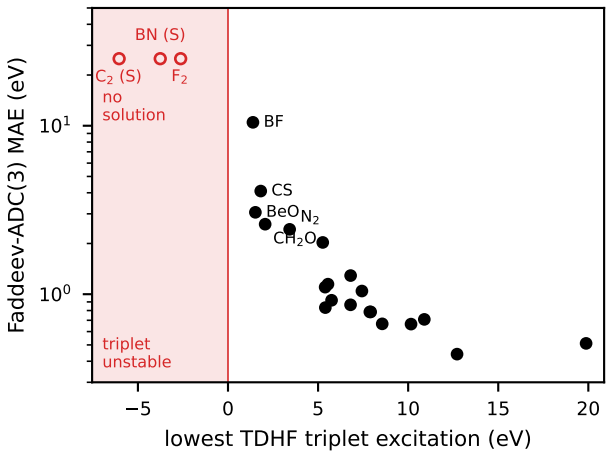


FIG. 6. Mean absolute error per molecule of Faddeev-ADC(3) (RPA pair channels) against sCI, versus the lowest TDHF triplet excitation energy of the RHF reference (aug-cc-pVDZ). For unstable references, where Faddeev-ADC(3) has no solution (open circles), the most negative eigenvalue of $A \pm B$ is shown; (S): also unstable in the singlet channel.

The observation that keeping it at first order (T1) or switching it off (T0) improves the results parallels the instabilities of RPA with exact exchange, which are removed by keeping exchange exact only at first order[131,132] and the improved stability of singlet-paired coupled cluster.[133,134]

This motivates the replacement of the uncoupled doubles amplitudes in the Faddeev-ADC(3) self-energy by CCD ones, which contain the couplings between the channels. The resulting method, to which we refer as CCD-Faddeev-ADC(3), is the same substitution that turns ADC(3) into CCD-ADC(3).[29] Since no RPA problem has to be solved, it is also defined in case of a triplet instability. The triplet problem disappears. The MAD over all 56 states with an sCI reference value is 0.23 eV, against 0.33 eV for ADC(3), 0.29 eV for CCD-ADC(3) and 0.18 eV for CCSD. The error distributions are shown in Fig. 5. For the subset of 48 states for which Faddeev-ADC(3) is actually defined, CCD-Faddeev-ADC(3) is a substantial improvement over Faddeev-ADC(3), with a MAD of 0.19 eV. This is very close to CCSD that comes with a MAD of 0.16 eV.

Maybe a bit surprisingly, the CCD amplitudes do not rescue FLEX. With a MAD of 1.20 eV, it is almost as bad as FLEX

with RPA channels (MAD of 1.42 eV). The CCD channels lower the IPs, which FLEX already underestimates. The deficiency of FLEX is not the quality of its pair channels but the way it combines them.

VII. CONCLUSIONS

Through the response function R, Eqs. (13) and (16), the self-energy is expressed in terms of the three-particle kernel. In the Faddeev form, Eq. (21), the only two-body input is the parquet vertex F. ADC(n) is the Schirmer matching of the Faddeev series at total order $n-2$ in v. The orders through which the various approximations are exact are collected in Table I.

Mapping the Faddeev form of parquet onto an effective Hamiltonian of the 2p1h (2h1p) space involves a few approximations. First, the parquet approximation, $\Lambda = \Lambda^{(1)}$, is combined with a static reference, $G = G_{\mathrm{HF}}$ in all lines of the vertex. Every building block of the vertex is then an instantaneous interaction $\bar{v}$ or a G_{HF} line, and the frequency integrals of the pair propagators turn into resolvents of configuration spaces (Sec. III B). Next, an own-channel approximation is invoked that consists in letting each pair scatter only through the ladder of its own channel with the bare kernel. Moreover, the ladders of different pairs do not overlap in time (Sec. III A).

In the own-channel and no-overlap approximation, Hermitian effective Hamiltonians (for a positive definite metric) are obtained that go beyond ADC by going beyond the TDA. At the ADC(3) level, this amounts to an effective Hamiltonian that replaces the first-order double amplitudes in the couplings with the ring-CCD (for the eh channel) and the ladder-CCD amplitudes (for the pp channel), and adds the metric $\mathcal{N}$ and the terms $B_\alpha T_\alpha$ to the matrix. Channel crossings that are not already part of the Faddeev kernel itself, as in the full CCD amplitude equations, are not considered. At first order, the Faddeev kernel does not contain any channel crossings.

Starting from the second-order Faddeev kernel, crossed channels already enter before any resummation, by Eq. (A8) and Eq. (24). These crossed ladders, together with the overlapping orderings of two first-order kernels and the line dressing, lead to the presence of 3p2h configurations. Obtaining an effective Hamiltonian requires including this configuration space, in the same way as in the ADC(4) approximation.

An advantage of our Faddeev construction is that it preserves the resolvent structure of the self-energy through a Hermitian effective Hamiltonian formalism, in the same way as in ADC. Unlike ADC, it also introduces a significantly larger amount of ground-state correlation through an infinite-order amplitude renormalization. The derivations shown here allow us to make precise connections to recent PSD completions of vertex-corrected self-energies.[33,125,126] The formal properties of the self-energies discussed here are summarized in Table VIII.

However, the numerical results of Faddeev-ADC(3) are very poor, which can be linked to the triplet channel. The same goes for the FLEX approximation and the PSDII approximation of Pavlyukh, Bruneval, and Förster[33]. Switching off the triplet channel entirely or leaving the triplet amplitudes at first order, like in ADC(3), significantly improves performance. Nevertheless, these schemes are clearly not competitive with methods based on the CC scheme, for instance diagonalization of the CCSD Hamiltonian. Clearly, coupling of the different correlation channels is mandatory whenever single-particle states are coupled to triplets beyond the TDA. To support this point, the replacement of the uncoupled amplitudes in Faddeev-ADC(3) with CCD ones results in improved results, but ADC(3) is only outperformed slightly. The performance is almost identical to CCD-ADC(3), and does not reach the performance of CCSD.

Appendix A: Parquet vertices

The Faddeev kernel is expanded order by order in $\bar{v}$,

$$\mathcal{K} = \mathcal{K}^{(1)} + \mathcal{K}^{(2)} + \mathcal{K}^{(3)} + O(v^4). \tag{A1}$$

At first order, all channels contain only the bare antisymmetrized interaction,

$$\begin{aligned} \Gamma^{\mathrm{pp},(1)} &= \Gamma^{\mathrm{eh},(1)} = \Gamma^{\overline{\mathrm{eh}},(1)} = \Lambda^{(1)} \equiv \Gamma^{(1)}, \\ \Gamma^{(1)}(12;1'2') &= -\mathrm{i}\,\bar{v}(12;1'2'), \\ \bar{v}(12;1'2') &\equiv v(12;1'2') - v(12;2'1') \,. \end{aligned} \tag{A2}$$

To all orders,

$$\begin{aligned} \Gamma^{\mathrm{eh}} &= \Lambda + \Phi^{\overline{\mathrm{eh}}} + \Phi^{\mathrm{pp}}, \\ \Gamma^{\overline{\mathrm{eh}}} &= \Lambda + \Phi^{\mathrm{eh}} + \Phi^{\mathrm{pp}}, \\ \Gamma^{\mathrm{pp}} &= \Lambda + \Phi^{\mathrm{eh}} + \Phi^{\overline{\mathrm{eh}}}, \\ F = \Gamma^{\lambda} + \Phi^{\lambda} &= \Lambda + \Phi^{\mathrm{pp}} + \Phi^{\mathrm{eh}} + \Phi^{\overline{\mathrm{eh}}} \,. \end{aligned} \tag{A3}$$

for each channel $\lambda \in \{\mathrm{pp}, \mathrm{eh}, \overline{\mathrm{eh}}\}$. Note that[35]

$$\Lambda = \Lambda^{(1)} + O(v^4) \,. \tag{A4}$$

Defining

$$\begin{aligned} (A \circ_{\mathrm{pp}} B)(12;1'2') &\equiv A(12;\bar{1}\bar{2})\, G(\bar{1}\bar{1}')\, G(\bar{2}\bar{2}') \\ &\quad \times B(\bar{1}'\bar{2}';1'2'), \end{aligned} \tag{A5}$$

$$\begin{aligned} (A \circ_{\mathrm{eh}} B)(12;1'2') &\equiv A(1\bar{4};1'\bar{3})\, G(\bar{3}\bar{3}')\, G(\bar{4}'\bar{4}) \\ &\quad \times B(\bar{3}'2;\bar{4}'2'), \end{aligned} \tag{A6}$$

the pp and eh BSEs are[4]

$$\begin{aligned} F &= \Gamma^{\mathrm{pp}} - \frac{1}{2}\,\Gamma^{\mathrm{pp}} \circ_{\mathrm{pp}} F \,, \\ F &= \Gamma^{\mathrm{eh}} + \Gamma^{\mathrm{eh}} \circ_{\mathrm{eh}} F \,. \end{aligned} \tag{A7}$$

At second order, the reducible vertices are

$$\begin{aligned} \Phi^{\mathrm{eh},(2)} &= \Gamma^{(1)} \circ_{\mathrm{eh}} \Gamma^{(1)}, \\ \Phi^{\mathrm{pp},(2)} &= -\frac{1}{2}\,\Gamma^{(1)} \circ_{\mathrm{pp}} \Gamma^{(1)}, \end{aligned} \tag{A8}$$

and at third order

$$\begin{aligned}\Phi^{\mathrm{eh},(3)} &= \Gamma^{\mathrm{eh},(2)} \circ_{\mathrm{eh}} \Gamma^{(1)} + \Gamma^{(1)} \circ_{\mathrm{eh}} \Gamma^{\mathrm{eh},(2)} \\ &\quad + \Gamma^{(1)} \circ_{\mathrm{eh}} \Gamma^{(1)} \circ_{\mathrm{eh}} \Gamma^{(1)}, \\ \Phi^{\mathrm{pp},(3)} &= -\frac{1}{2}(\Gamma^{\mathrm{pp},(2)} \circ_{\mathrm{pp}} \Gamma^{(1)} + \Gamma^{(1)} \circ_{\mathrm{pp}} \Gamma^{\mathrm{pp},(2)}) \\ &\quad + \frac{1}{4}\,\Gamma^{(1)} \circ_{\mathrm{pp}} \Gamma^{(1)} \circ_{\mathrm{pp}} \Gamma^{(1)}.\end{aligned} \tag{A9}$$

Together with Eqs. (A3) and (A4) and fermionic crossing,

$$\Phi^{\overline{\mathrm{eh}}}(12;1'2') = -\Phi^{\mathrm{eh}}(21;1'2')\,, \tag{A10}$$

this gives $\Gamma^{\lambda,(2)}$ and $\Gamma^{\lambda,(3)}$.

Appendix B: Pair and three-particle propagators

Here we derive the propagators stated in Sec. III B.

1. Notation

We use real canonical spin-orbitals and map numbers to orbitals with $1,2,1',2' \to p,q,r,s$. Since $\bar{v}$, Eq. (A2), is instantaneous,

$$\bar{v}(12;1'2') \to \langle pq||rs\rangle\ \delta(t_1-t_2)\,\delta(t_1-t_{1'})\,\delta(t_2-t_{2'}), \tag{B1}$$

with

$$\langle pq||rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle\ . \tag{B2}$$

The antisymmetrizer of a pair of indices is

$$\hat{P}^-_{ab} = 1 - \hat{P}_{ab}\,, \tag{B3}$$

where $\hat{P}_{ab}$ exchanges a and b. In canonical spin-orbitals p with energies ϵ_p, occupations n_p ($n_i = 1$, $n_a = 0$, $\bar{n}_p = 1 - n_p$), and $\sigma_p = \bar{n}_p - n_p$, the HF Green's function is

$$G_{\mathrm{HF},p}(\omega) = \frac{\bar{n}_p}{\omega - \epsilon_p + \mathrm{i}0^+} + \frac{n_p}{\omega - \epsilon_p - \mathrm{i}0^+}. \tag{B4}$$

2. Pair problems and reducible vertices

In the own-channel approximation, each pair scatters with the bare kernel. In the eh channel this is the TDHF problem, in the pp channel the ppRPA problem,

$$\eta^\lambda \mathbf{H}^\lambda Z_n = \Omega_n Z_n\,, \qquad \eta^\lambda = \mathrm{diag}(\mathbf{1},-\mathbf{1})\,, \tag{B5}$$

with

$$\mathbf{H}^{\mathrm{eh}} = \begin{pmatrix} A^{\mathrm{eh}} & B^{\mathrm{eh}} \\ B^{\mathrm{eh}} & A^{\mathrm{eh}} \end{pmatrix}, \tag{B6}$$

$$\begin{aligned} A^{\mathrm{eh}}_{ia,jb} &= (\epsilon_a - \epsilon_i)\,\delta_{ab}\delta_{ij} + \langle aj||ib\rangle\,, \\ B^{\mathrm{eh}}_{ia,jb} &= \langle ab||ij\rangle\ , \end{aligned}$$

and

$$\mathbf{H}^{\mathrm{pp}} = \begin{pmatrix} A^{\mathrm{pp}} & B^{\mathrm{pp}} \\ B^{\mathrm{pp}\dagger} & D^{\mathrm{pp}} \end{pmatrix}, \tag{B7}$$

$$\begin{aligned} A^{\mathrm{pp}}_{cd,ef} &= (\epsilon_c + \epsilon_d)\,\delta_{ce}\delta_{df} + \langle cd||ef\rangle\,, \\ B^{\mathrm{pp}}_{cd,kl} &= \langle cd||kl\rangle\,, \\ D^{\mathrm{pp}}_{kl,mn} &= -(\epsilon_k + \epsilon_l)\,\delta_{km}\delta_{ln} + \langle kl||mn\rangle\ , \end{aligned}$$

with $c<d$, $k<l$. The eh eigenvalues come in pairs $\pm\Omega_n$, with eigenvectors $Z_n = (X_n, Y_n)$ and $\tilde{Z}_n = (Y_n, X_n)$. The pp spectrum is not symmetric: the positive-norm eigenvectors are the $N+2$ states (Ω^+_n), the negative-norm ones the $N-2$ states (Ω^-_m).

The pair propagators give the reducible vertices of the own-channel approximation, the resummed counterparts of $\Phi^{\lambda,(2)}$ in Eq. (A8), as sums over pair excitations with the two-body pair vertices M^{eh} and M^{pp},

$$\begin{aligned} \mathrm{i}\Phi^{\mathrm{eh}}_{pqrs}(\Omega) &= \sum_{\Omega_n>0}\left[\frac{M^{\mathrm{eh}}_{pr,n}\,M^{\mathrm{eh}}_{qs,n}}{\Omega - \Omega_n + \mathrm{i}0^+} - \frac{M^{\mathrm{eh}}_{rp,n}\,M^{\mathrm{eh}}_{sq,n}}{\Omega + \Omega_n - \mathrm{i}0^+}\right], \\ M^{\mathrm{eh}}_{pr,n} &= \sum_{ia}\left[\langle pa||ri\rangle\ X^n_{ia} + \langle pi||ra\rangle\ Y^n_{ia}\right], \end{aligned} \tag{B8}$$

and

$$\begin{aligned} \mathrm{i}\Phi^{\mathrm{pp}}_{pqrs}(\Omega) &= \sum_{n\in N+2}\frac{M^{\mathrm{pp}}_{pq,n}\,M^{\mathrm{pp}}_{rs,n}}{\Omega - \Omega^+_n + \mathrm{i}0^+} - \sum_{m\in N-2}\frac{M^{\mathrm{pp}}_{pq,m}\,M^{\mathrm{pp}}_{rs,m}}{\Omega - \Omega^-_m - \mathrm{i}0^+}, \\ M^{\mathrm{pp}}_{pq,n} &= \sum_{c<d}\langle pq||cd\rangle\ X^n_{cd} + \sum_{k<l}\langle pq||kl\rangle\ Y^n_{kl}\,. \end{aligned} \tag{B9}$$

For real orbitals the backward eh vertex is the forward one with exchanged orbital indices, $M^{\mathrm{eh}}_{rp,n}$. The same first-order vertex appears in the vertex-corrected GW self-energies of Refs. 32,74,77. In the eh variables, Φ^{eh} depends only on the bosonic transfer frequency Ω; the crossed channel, $\overline{\mathrm{eh}}$, follows by crossing, Eq. (A10).

3. Pair propagators

The non-interacting particle–particle propagator is

$$L^{\mathrm{pp}}_0(12;1'2') = G(11')G(22') - G(12')G(21')\,, \tag{B10}$$

and performing the frequency integrations,

$$\begin{aligned} L^{\mathrm{pp}}_{0,xy}(\Omega) &= \int\frac{\mathrm{d}\nu}{2\pi}\,G_x(\nu)\,G_y(\Omega-\nu) \\ &= \frac{-\mathrm{i}\sigma_x\,\delta_{\sigma_x\sigma_y}}{\Omega - (\epsilon_x+\epsilon_y) + \mathrm{i}\sigma_x 0^+}\ , \end{aligned} \tag{B11}$$

with its indices restricted to $(xy) \in \{(cd),\, c<d\} \cup \{(kl),\, k<l\}$. The particle–hole propagator is

$$L^{\mathrm{eh}}_0(31;1'3') = G(33')G(11')\,. \tag{B12}$$

After performing the frequency integrations, we obtain

$$L^{\mathrm{eh}}_{0,xr}(\Omega) = \int \frac{\mathrm{d}\nu}{2\pi} G_x(\nu)\, G_r(\nu+\Omega) = \frac{\mathrm{i}\sigma_r\, \delta_{\sigma_x,-\sigma_r}}{\Omega - (\epsilon_r - \epsilon_x) + \mathrm{i}\sigma_r 0^+} , \tag{B13}$$

with its indices restricted to $(xr) \in \{(ia)\} \cup \{(ai)\}$.

The interacting propagators follow from the own-channel BSEs, Eq. (25). Restricting the pp pairs to $c < d$, $k < l$ absorbs the $\frac{1}{2}$ of the pp BSE, and $\mathrm{i}\Gamma^{(1)}_{pqrs} = \langle pq \| rs \rangle$ by Eqs. (A2) and (B1). With the pair Hamiltonians of Eqs. (B6) and (B7),

$$L^{\mathrm{eh}}(\Omega) = \mathrm{i}(\Omega\,\eta^{\mathrm{eh}} - \mathbf{H}^{\mathrm{eh}})^{-1} ,$$
$$L^{\mathrm{pp}}(\Omega) = -\mathrm{i}(\Omega\,\eta^{\mathrm{pp}} - \mathbf{H}^{\mathrm{pp}})^{-1} ,$$

with the spectral representations

$$L^{\mathrm{eh}}(\Omega) = \mathrm{i}\sum_{\Omega_n>0}\left[\frac{Z_n Z_n^\dagger}{\Omega - \Omega_n + \mathrm{i}0^+} - \frac{\tilde{Z}_n \tilde{Z}_n^\dagger}{\Omega + \Omega_n - \mathrm{i}0^+}\right] \tag{B14}$$

and

$$L^{\mathrm{pp}}(\Omega) = -\mathrm{i}\sum_{n\in N+2}\frac{Z_n Z_n^\dagger}{\Omega - \Omega_n^+ + \mathrm{i}0^+} + \mathrm{i}\sum_{m\in N-2}\frac{Z_m Z_m^\dagger}{\Omega - \Omega_m^- - \mathrm{i}0^+} . \tag{B15}$$

Inserting them into the own-channel reducible vertices gives Eqs. (B8) and (B9).

4. Three-particle propagators

For the pp pair, Eqs. (15), (17) and (B10) give

$$\mathcal{G} + \mathcal{K}_{\mathrm{pp}}[F^{\mathrm{pp}}_{\mathrm{oc}}]\mathcal{G} = G(33')\Big[L^{\mathrm{pp}}_0 - \tfrac{1}{4}\, L^{\mathrm{pp}}_0 F^{\mathrm{pp}}_{\mathrm{oc}} L^{\mathrm{pp}}_0\Big](12;1'2') = G(33')\, L^{\mathrm{pp}}(12;1'2') , \tag{B16}$$

Working out the frequency integral with Eqs. (B4) and (B15), with r the spectator,

$$Q_{\mathrm{pp};xyr,uwr}(\omega) = \int \frac{\mathrm{d}\Omega}{2\pi} L^{\mathrm{pp}}_{xy,uw}(\Omega)\, G_r(\Omega-\omega) = n_r \sum_{n\in N+2}\frac{Z^n_{xy} Z^n_{uw}}{\omega + \epsilon_r - \Omega_n^+ + \mathrm{i}0^+} + \bar{n}_r \sum_{m\in N-2}\frac{Z^m_{xy} Z^m_{uw}}{\omega + \epsilon_r - \Omega_m^- - \mathrm{i}0^+} .$$

which is Eq. (53).

For the eh pair, Eqs. (15), (17) and (B12) give

$$\mathcal{G} + \mathcal{K}_{\mathrm{eh}}[F^{\mathrm{eh}}_{\mathrm{oc}}]\mathcal{G} = G(22')\Big[L^{\mathrm{eh}}_0 + L^{\mathrm{eh}}_0 F^{\mathrm{eh}}_{\mathrm{oc}} L^{\mathrm{eh}}_0\Big](31;1'3') - G(21')\Big[L^{\mathrm{eh}}_0 + L^{\mathrm{eh}}_0 F^{\mathrm{eh}}_{\mathrm{oc}} L^{\mathrm{eh}}_0\Big](31;2'3') = G(22')\, L^{\mathrm{eh}}(31;1'3') - G(21')\, L^{\mathrm{eh}}(31;2'3') , \tag{B17}$$

The propagator of the $\overline{\mathrm{eh}}$ pair follows with $1 \leftrightarrow 2$. Unlike the pp pair, the eh pair is not antisymmetrized in $\mathcal{G}$, so that the exchange of the lines 1 and 2 appears as the second term. Working out the frequency integral with Eqs. (B4) and (B14),

$$Q_{\mathrm{eh};xrs,yus}(\omega) = \int \frac{\mathrm{d}\Omega}{2\pi} L^{\mathrm{eh}}_{xr,yu}(\Omega)\, G_s(\omega-\Omega) = \bar{n}_s \sum_{\Omega_n>0}\frac{Z^n_{xr} Z^n_{yu}}{\omega - \epsilon_s - \Omega_n + \mathrm{i}0^+} + n_s \sum_{\Omega_n>0}\frac{\tilde{Z}^n_{xr} \tilde{Z}^n_{yu}}{\omega - \epsilon_s + \Omega_n - \mathrm{i}0^+} .$$

which is Eq. (54). With L^{pp}_0 in place of L^{pp}, Eq. (53) gives the free propagation $\mathcal{G}$, Eq. (46).

Appendix C: Riccati equations and decoupling conditions

Consider a space $\mathcal{P} \oplus \mathcal{Q}$, where $\mathcal{P}$ is some active space. Let $x \in \mathcal{P} \oplus \mathcal{Q}$ be an eigenvector of H and write

$$x = \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} X \\ TX \end{pmatrix} \tag{C1}$$

which implies $Y = TX$. We then define the graph of T as

$$\mathcal{P}_T = \mathrm{graph}(T) = \left\{ \begin{pmatrix} X \\ TX \end{pmatrix} , X \in \mathcal{P} \right\} \tag{C2}$$

and require it to be invariant under the action of the partitioned Hamiltonian

$$\begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix}\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix}\begin{pmatrix} X \\ TX \end{pmatrix} = \begin{pmatrix} (H_{PP} + H_{PQ}T)\, X \\ (H_{QP} + H_{QQ}T)\, X \end{pmatrix} . \tag{C3}$$

Invariance under the action of H then gives the condition

$$(H_{QP} + H_{QQ}T)\, X = T\,(H_{PP} + H_{PQ}T)\, X , \tag{C4}$$

and as this must be valid for all $X \in \mathcal{P}$, this yields the Riccati equation

$$H_{QP} + H_{QQ}T = T H_{PP} + T H_{PQ} T . \tag{C5}$$

The rewriting of the pp and eh-RPA equations, Eqs. (B7) and (B6), as the Riccati equations, Eq. (57), is easily seen to be a special case of this general transformation. Equivalently, with

$$\Theta = \begin{pmatrix} \mathbf{1} & 0 \\ T & \mathbf{1} \end{pmatrix} , \qquad \Theta^{-1} = \begin{pmatrix} \mathbf{1} & 0 \\ -T & \mathbf{1} \end{pmatrix} , \tag{C6}$$

whose first block column spans $\mathcal{P}_T$,

$$\Theta^{-1} H \Theta = \begin{pmatrix} H_{PP} + H_{PQ}T & H_{PQ} \\ H_{QP} + H_{QQ}T - T H_{PP} - T H_{PQ} T & H_{QQ} - T H_{PQ} \end{pmatrix} . \tag{C7}$$

The lower-left block is the Riccati equation, and the invariance of $\mathcal{P}_T$ is the decoupling condition

$$Q\,\Theta^{-1} H \Theta\, P = 0 , \tag{C8}$$

with P and Q the projectors onto $\mathcal{P}$ and $\mathcal{Q}$. $\Theta^{-1}H\Theta$ is then block upper triangular, and $H_{PP} + H_{PQ}T$ carries the eigenvalues of H on $\mathcal{P}_T$. For $H = \eta^\lambda \mathbf{H}^\lambda$ of Eqs. (B6) and (B7), *i.e.* $H_{PP} = A_\alpha$, $H_{PQ} = B_\alpha$, $H_{QP} = -B_\alpha^\dagger$ and $H_{QQ} = -D_\alpha$, the lower-left block is minus the left-hand side of Eq. (57).

As an example, consider the wave operator

$$\mathcal{W} = \Theta P = P + T \;, \qquad P\mathcal{W} = P \;, \tag{C9}$$

which maps $\mathcal{P}$ onto $\mathcal{P}_T$. The invariance of $\mathcal{P}_T$ is then the Bloch equation[135–137]

$$H\mathcal{W} = \mathcal{W}H\mathcal{W} \;, \tag{C10}$$

whose QP block is the Riccati equation and whose PP block is an identity. The effective Hamiltonian is the upper-left block,

$$H_T = PH\mathcal{W} = P\Theta^{-1}H\Theta P = H_{PP} + H_{PQ}T \;. \tag{C11}$$

For $H = H_0 + V$ with $[H_0, P] = 0$, Eq. (C10) becomes the generalized Bloch equation

$$[\mathcal{W}, H_0] = V\mathcal{W} - \mathcal{W}PV\mathcal{W} \;, \tag{C12}$$

the starting point of Rayleigh–Schrödinger perturbation theory for $\mathcal{W}$. For a one-dimensional $\mathcal{P}$ spanned by Φ, T contains the CI coefficients of $\Psi = \mathcal{W}\Phi$ in intermediate normalization, $H_T = E$, and Eq. (C10) is the Schrödinger equation projected onto $\mathcal{Q}$.

1. Comparison to Löwdin downfolding

The Bloch equation should be compared with Löwdin downfolding, which generates the frequency-dependent effective Hamiltonian acting in $\mathcal{P}$-space only,

$$H_{\mathrm{L}}(\omega) = H_{PP} + H_{PQ}\,(\omega - H_{QQ})^{-1}\,H_{QP} \;, \tag{C13}$$

Comparison with Eq. (C11) then shows that

$$T(\omega) = (\omega - H_{QQ})^{-1}\,H_{QP} \;. \tag{C14}$$

$H_{\mathrm{L}}(\omega)$ is Hermitian for Hermitian H and has the form of a self-energy, and $\det(\omega - H_{\mathrm{L}}(\omega)) = 0$ yields all eigenvalues of H with a non-vanishing $\mathcal{P}$ component, but it is energy-dependent and has to be solved root by root. The Bloch amplitude is instead a single, energy-independent map for the $\dim\mathcal{P}$ selected roots,

$$T = (T(\omega_1)x_1, \ldots, T(\omega_{\dim\mathcal{P}})x_{\dim\mathcal{P}})X^{-1} \;, \tag{C15}$$

at the price of a nonlinear equation, a non-Hermitian H_T, and one solution for each choice of roots. Expanded in V, Eq. (C13) gives Brillouin–Wigner and Eq. (C10) Rayleigh–Schrödinger perturbation theory. The two are related by a Taylor expansion of $T(\omega_k)$ around a reference energy E_0: with $g_0 = (E_0 - H_{QQ})^{-1}$ and $\omega_k x_k = H_T x_k$,

$$T = \sum_{n\geq 0} g_0^{n+1} H_{QP}\,(E_0 - H_T)^n \;, \tag{C16}$$

i.e. the energy dependence of $T(\omega)$ becomes the dependence on H_T, which is the same for all selected roots.

Appendix D: The ADC hierarchy

In this appendix, we work out ADC(3) explicitly, as an illustrative case of Sec. II D. ADC(1), *i.e.* HF, needs no three-particle propagator at all, and ADC(2), *i.e.* GF2, only the 2p1h propagator $\mathcal{G}$.

1. ADC(3) from the first-order Faddeev kernel

We here derive the ADC(3) equation starting from Eq. (13) with $R_{\mathrm{ADC(3)}}$ of Eq. (24), *i.e.* with the first-order Faddeev kernel acting once on $\mathcal{G}$. We then collect all diagrams by the pair on which $\mathcal{K}^{(1)}$ acts, *i.e.* in the pp and eh channels. This will give us the GF3 self-energy. Then we use Schirmer's approach of imposing the resolvent form of the self-energy and diagrammatic matching to get ADC(3).

The self-energy to third order is

$$\Sigma - \Sigma_{\mathrm{HF}}[G_{\mathrm{HF}}] = \Sigma^{(2)} + \Sigma^{(3)} + O(v^4) \;. \tag{D1}$$

The Green's function is expanded to second order, which will later give the static self-energy to third order (notice that the $G_{\mathrm{HF}}\Sigma_{\mathrm{HF}}G_{\mathrm{HF}}\Sigma_{\mathrm{HF}}G_{\mathrm{HF}}$ term has been absorbed in the mean field, $G^{-1} = G_{\mathrm{HF}}^{-1} + O(v^2)$),

$$G = G_{\mathrm{HF}} + G_{\mathrm{HF}}\,\Sigma^{(2)}G_{\mathrm{HF}} + O(v^3) \;, \tag{D2}$$

$\mathcal{G}$ is built from G_{HF},

$$\mathcal{G} = \mathcal{G}_{\mathrm{HF}} + O(v^2) \;. \tag{D3}$$

For ADC(3), we must consider the first-order Faddeev kernel,

$$\mathcal{K} = \mathcal{K}^{(1)} + O(v^2) \;, \tag{D4}$$

that only contains the bare antisymmetrized interactions in all channels, Eq. (A2). Recall that ADC(3) follows from the second line of Eq. (24), where

$$\sum_\alpha \mathcal{K}_\alpha[F^{(1)}] = \mathcal{K}^{(1)} \;.$$

The third-order self-energy splits into static and dynamical contributions,

$$\Sigma^{(3)}(11') = \Sigma^{(3)}_\infty(11') + \Sigma^{(3)}_{\mathcal{K}}(11'), \tag{D5}$$

$$\begin{aligned}\Sigma^{(3)}_\infty(11') = &-\mathrm{i}[v(1\bar{2};1'\bar{2}') - v(1\bar{2};\bar{2}'1')] \\ &\times [G_{\mathrm{HF}}\Sigma^{(2)}G_{\mathrm{HF}}](\bar{2}'\bar{2}^+),\end{aligned} \tag{D6}$$

$$\begin{aligned}\Sigma^{(3)}_{\mathcal{K}}(11') = &\; v(1\bar{2};\bar{3}'\bar{2}')\,G_{\mathrm{HF}}(\bar{3}'\bar{4})\,v(\bar{4}\bar{5};\bar{6}'\bar{7}') \\ &\times [\mathcal{K}^{(1)}\mathcal{G}_{\mathrm{HF}}](\bar{2}'\bar{6}'\bar{7}';\bar{3}\bar{2}\bar{5}^+) \\ &\times G^{-1}_{\mathrm{HF}}(\bar{3}1').\end{aligned} \tag{D7}$$

Eq. (D6) is the HF self-energy evaluated with the second-order density matrix,

$$\Sigma^{(3)}_{\infty,pq} = \sum_{rs} \langle pr\|qs\rangle\; \gamma^{(2)}_{sr}. \tag{D8}$$

Defining the first-order doubles

$$t_{ij}^{ab} = \frac{\langle ab||ij\rangle}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} , \tag{D9}$$

and second-order singles amplitudes

$$t_i^{a(2)} = \frac{1}{\epsilon_i - \epsilon_a}\Big[\frac{1}{2}\sum_{kbc} t_{ik}^{bc}\,\langle bc||ak\rangle - \frac{1}{2}\sum_{jkc} t_{jk}^{ac}\,\langle ic||jk\rangle\Big] , \tag{D10}$$

the blocks of $\gamma^{(2)}$ are[28]

$$\begin{aligned}
\gamma_{ij}^{(2)} &= -\frac{1}{2}\sum_{kab} t_{ik}^{ab} t_{jk}^{ab} , \\
\gamma_{ab}^{(2)} &= \frac{1}{2}\sum_{ijc} t_{ij}^{ac} t_{ij}^{bc} , \\
\gamma_{ia}^{(2)} &= \gamma_{ai}^{(2)} = t_i^{a(2)} .
\end{aligned} \tag{D11}$$

Using Eqs. (16) and (17) in Eq. (D7) with $\chi = \mathcal{G}$, we obtain the contributions of the three channels to $\Sigma^{(3)}_{\mathcal{K}}$. Here and below, $G \equiv G_{\rm HF}$, and all interactions appear antisymmetrized, Eq. (A2). For the pp channel, we get

$$\begin{aligned}
\Sigma^{\rm pp}_{\mathcal{K}}(11') = \frac{\mathrm{i}}{4}\,&\bar{v}(1\bar{2};\bar{3}'\bar{2}')\,G(\bar{3}'\bar{4})\,G(\bar{2}'\bar{5}^+) \\
&\times \bar{v}(\bar{4}\bar{5};\bar{6}'\bar{7}')\,G(\bar{6}'\bar{8})\,G(\bar{7}'\bar{9}) \\
&\times \bar{v}(\bar{8}\bar{9};1'\bar{9}')\,G(\bar{9}'\bar{2}) .
\end{aligned} \tag{D12}$$

For the eh, $\overline{\rm eh}$ channels, we obtain

$$\begin{aligned}
\Sigma^{\rm eh}_{\mathcal{K}}(11') = -\frac{\mathrm{i}}{2}\,&\bar{v}(1\bar{2};\bar{3}'\bar{2}')\,G(\bar{3}'\bar{4})\,\bar{v}(\bar{4}\bar{5};\bar{6}'\bar{7}') \\
&\times G(\bar{2}'\bar{8})\,G(\bar{6}'\bar{9})\,\bar{v}(\bar{8}\bar{9};1'\bar{8}') \\
&\times G(\bar{8}'\bar{5}^+)\,G(\bar{7}'\bar{2}) = \Sigma^{\overline{\rm eh}}_{\mathcal{K}}(11') .
\end{aligned} \tag{D13}$$

In the time domain, the non-interacting pair propagators follow from $G_{{\rm HF},p}(t) = -\mathrm{i}\sigma_p\,\theta(\sigma_p t)\,e^{-\mathrm{i}\epsilon_p t}$,

$$L^{\rm pp}_{0,xy}(t) = G_x(t)\,G_y(t) = -\delta_{\sigma_x\sigma_y}\,\theta(\sigma_x t)\,e^{-\mathrm{i}(\epsilon_x+\epsilon_y)t} , \tag{D14}$$

and

$$L^{\rm eh}_{0,xr}(t) = G_x(-t)\,G_r(t) = \delta_{\sigma_x,-\sigma_r}\,\theta(\sigma_r t)\,e^{-\mathrm{i}(\epsilon_r-\epsilon_x)t} . \tag{D15}$$

Exploiting the temporal structure of $\bar{v}$ and the space-time to orbital index mapping of Eq. (B1), and with Eq. (D14) the pp term becomes

$$\begin{aligned}
\Sigma^{\rm pp}_{\mathcal{K},pq}(t_1,t_{1'}) = \frac{\mathrm{i}}{4}&\sum_{rxyuw}\langle pr||xy\rangle\,\langle xy||uw\rangle\,\langle uw||qr\rangle \\
&\times\int \mathrm{d}\tau\; L^{\rm pp}_{0,xy}(t_1-\tau)\,L^{\rm pp}_{0,uw}(\tau-t_{1'}) \\
&\times G_r(t_{1'}-t_1) ,
\end{aligned} \tag{D16}$$

in the orbital basis. Summing up the identical eh and $\overline{\rm eh}$ channels and with Eq. (D15), the sum of $\Sigma^{\rm eh}$ and $\Sigma^{\overline{\rm eh}}$ becomes

$$\begin{aligned}
\Sigma^{{\rm eh}+\overline{\rm eh}}_{\mathcal{K},pq}(t_1,t_{1'}) = -\mathrm{i}&\sum_{rsxyu}\langle pr||xy\rangle\,\langle xs||ur\rangle\,\langle yu||qs\rangle \\
&\times\int \mathrm{d}\tau\; L^{\rm eh}_{0,rx}(t_1-\tau)\,L^{\rm eh}_{0,su}(\tau-t_{1'}) \\
&\times G_y(t_1-t_{1'}) .
\end{aligned} \tag{D17}$$

Not coincidentally, one can identify the second-order pp and eh reducible vertices, Eq. (A8), of the parquet equations.

a. GF3 self-energy

We perform the integrals over τ by regarding each space-time term as a circuit and using energy conservation at the intermediate vertices, which leaves contour integrals over the channel frequency. For the pp channel, one then gets

$$\begin{aligned}
\Sigma^{\rm pp}_{\mathcal{K},pq}(\omega) = \frac{\mathrm{i}}{4}&\sum_{rxyuw}\langle pr||xy\rangle\,\langle xy||uw\rangle\,\langle uw||qr\rangle \\
&\times\int\frac{\mathrm{d}\Omega}{2\pi}\,L^{\rm pp}_{0,xy}(\Omega)\,L^{\rm pp}_{0,uw}(\Omega)\,G_r(\Omega-\omega),
\end{aligned} \tag{D18}$$

with Eq. (B11). In the eh + $\overline{\rm eh}$ channel, Eq. (D17), we obtain

$$\begin{aligned}
\Sigma^{{\rm eh}+\overline{\rm eh}}_{\mathcal{K},pq}(\omega) = -\mathrm{i}&\sum_{rsxyu}\langle pr||xy\rangle\,\langle xs||ur\rangle\,\langle yu||qs\rangle \\
&\times\int\frac{\mathrm{d}\Omega}{2\pi}\,L^{\rm eh}_{0,rx}(\Omega)\,L^{\rm eh}_{0,su}(\Omega)\,G_y(\omega-\Omega),
\end{aligned} \tag{D19}$$

with Eq. (B13). In both channels, the kernel factorizes into two one-frequency pair propagators in the channel transfer frequency Ω; only the Ω integral couples them to the spectator line (r in pp, y in eh). What remains is to perform the contour integrals over Ω. For the pp channel, this gives

$$\Sigma^{\rm pp}_{\mathcal{K},pq}(\omega) = \frac{1}{4}\sum_{abcdi}\frac{\langle pi||ab\rangle\,\langle ab||cd\rangle\,\langle cd||qi\rangle}{(\omega-\Delta_{abi})(\omega-\Delta_{cdi})} \tag{D20}$$

$$+\frac{1}{4}\sum_{abkli}\frac{\langle pi||ab\rangle\,t_{kl}^{ab}\,\langle kl||qi\rangle}{\omega-\Delta_{abi}} \tag{D21}$$

$$+\frac{1}{4}\sum_{cdkli}\frac{\langle pi||kl\rangle\,t_{kl}^{cd}\,\langle cd||qi\rangle}{\omega-\Delta_{cdi}} \tag{D22}$$

$$-\frac{1}{4}\sum_{ijkla}\frac{\langle pa||ij\rangle\,\langle ij||kl\rangle\,\langle kl||qa\rangle}{(\omega-\Delta_{ija})(\omega-\Delta_{kla})} \tag{D23}$$

$$+\frac{1}{4}\sum_{bckla}\frac{\langle pa||bc\rangle\,t_{kl}^{bc}\,\langle kl||qa\rangle}{\omega-\Delta_{kla}} \tag{D24}$$

$$+\frac{1}{4}\sum_{ijbca}\frac{\langle pa||ij\rangle\,t_{ij}^{bc}\,\langle bc||qa\rangle}{\omega-\Delta_{ija}} . \tag{D25}$$

For the eh + $\overline{\text{eh}}$ channel,

$$\Sigma^{\text{eh+}\overline{\text{eh}}}_{\mathcal{K},pq}(\omega) = \sum_{abcij} \frac{\langle pi||ab\rangle\,\langle aj||ci\rangle\,\langle bc||qj\rangle}{(\omega-\Delta_{abi})(\omega-\Delta_{bcj})} \tag{D26}$$

$$- \sum_{abcik} \frac{\langle pi||ab\rangle\; t^{ac}_{ik}\;\langle bk||qc\rangle}{\omega-\Delta_{abi}} \tag{D27}$$

$$+ \sum_{abcjk} \frac{\langle pc||ja\rangle\; t^{bc}_{jk}\;\langle ab||qk\rangle}{\omega-\Delta_{abk}} \tag{D28}$$

$$- \sum_{ijkab} \frac{\langle pb||ij\rangle\,\langle ia||kb\rangle\,\langle jk||qa\rangle}{(\omega-\Delta_{ijb})(\omega-\Delta_{jka})} \tag{D29}$$

$$- \sum_{ijkab} \frac{\langle pj||ai\rangle\; t^{ab}_{jk}\;\langle ik||qb\rangle}{\omega-\Delta_{ikb}} \tag{D30}$$

$$+ \sum_{ijkab} \frac{\langle pb||ij\rangle\; t^{ab}_{ik}\;\langle ja||qk\rangle}{\omega-\Delta_{ijb}}\,. \tag{D31}$$

We have used the shorthand of Eq. (47). These terms constitute the complete dynamic GF3 self-energy, which is for instance given in the appendix of Ref. 21.

b. *From GF3 to ADC(3)*

What remains is to match the GF3 terms with the ADC(3) ones, following the method introduced by Schirmer.[11] The procedure is rather simple: we impose the resolvent structure on the self-energy

$$\Sigma_{\text{ADC(3)}}(\omega) = \Sigma(\infty) + U^{\text{II}}(\omega - K^{\text{II}} - C^{\text{II}})^{-1}U^{\text{II}\dagger} + U^{\text{I}}(\omega + K^{\text{I}} + C^{\text{I}})^{-1}U^{\text{I}\dagger}, \tag{D32}$$

with the static self-energy $\Sigma(\infty) = \Sigma_{\text{HF}} + \Sigma^{(3)}_{\infty}$ and

$$U^{\text{II}} = U^{\text{II},(1)} + U^{\text{II},(2)}\,, \qquad U^{\text{I}} = U^{\text{I},(1)} + U^{\text{I},(2)}\,. \tag{D33}$$

All terms are then classified according to their number of dynamical energy denominators. The ones that only contain one go inside U, and the ones that contain two go inside C. Terms with the dynamical denominator $\omega - \Delta_{abi}$ (2p1h, $N+1$) belong to block II, terms with $\omega - \Delta_{ija}$ (2h1p, $N-1$) to block I. We use the zeroth-order matrices and first-order couplings of Eq. (49). The unrestricted sums of the GF3 terms are converted with

$$\sum_{a<b}\sum_{c<d} x_{ab}\,\hat{P}^{-}_{ab}\hat{P}^{-}_{cd} f_{ab,cd}\,y_{cd} = \sum_{abcd} x_{ab}\,f_{ab,cd}\,y_{cd},$$
$$\sum_{a<b} x_{ab}\,\hat{P}^{-}_{ab} g_{ab} = \sum_{ab} x_{ab}\,g_{ab}, \tag{D34}$$

for x, y antisymmetric in their pair indices, with $\hat{P}^{-}_{ab}$ of Eq. (B3).

In block II, the two-denominator terms, Eqs. (D20) and (D26), combine into

$$\Sigma^{\text{II}}_{C}(\omega) = U^{\text{II},(1)}(\omega - K^{\text{II}})^{-1}C^{\text{II}}(\omega - K^{\text{II}})^{-1}U^{\text{II},(1)\dagger},$$
$$C^{\text{II}}_{abi,cdj} = \langle ab||cd\rangle\,\delta_{ij} - \hat{P}^{-}_{ab}\hat{P}^{-}_{cd}\,\langle aj||ci\rangle\,\delta_{bd}, \tag{D35}$$

where we used $\langle bc||qj\rangle = -\langle qj||cb\rangle$. The one-denominator terms, Eqs. (D21), (D22), (D27) and (D28), give

$$\Sigma^{\text{II}}_{U}(\omega) = U^{\text{II},(2)}(\omega - K^{\text{II}})^{-1}U^{\text{II},(1)\dagger} + U^{\text{II},(1)}(\omega - K^{\text{II}})^{-1}U^{\text{II},(2)\dagger},$$
$$U^{\text{II},(2)}_{p;abi} = \frac{1}{2}\sum_{jl} t^{ab}_{jl}\,\langle pi||jl\rangle - \hat{P}^{-}_{ab}\sum_{kc} t^{ac}_{ki}\,\langle pc||kb\rangle\,, \tag{D36}$$

where we used $t^{ac}_{ik}\,\langle bk||qc\rangle = t^{ac}_{ki}\,\langle qc||kb\rangle$.

In block I, the two-denominator terms, Eqs. (D23) and (D29), combine into

$$\Sigma^{\text{I}}_{C}(\omega) = -U^{\text{I},(1)}(\omega + K^{\text{I}})^{-1}C^{\text{I}}(\omega + K^{\text{I}})^{-1}U^{\text{I},(1)\dagger},$$
$$C^{\text{I}}_{ija,klb} = \langle ij||kl\rangle\,\delta_{ab} - \hat{P}^{-}_{ij}\hat{P}^{-}_{kl}\,\langle ib||ka\rangle\,\delta_{jl}, \tag{D37}$$

where we used $\langle jk||qa\rangle = -\langle qa||kj\rangle$. The sign is such that Σ^{I}_{C} is the first-order term of $U^{\text{I},(1)}(\omega + K^{\text{I}} + C^{\text{I}})^{-1}U^{\text{I},(1)\dagger}$. The one-denominator terms, Eqs. (D24), (D25), (D30) and (D31), give

$$\Sigma^{\text{I}}_{U}(\omega) = U^{\text{I},(2)}(\omega + K^{\text{I}})^{-1}U^{\text{I},(1)\dagger} + U^{\text{I},(1)}(\omega + K^{\text{I}})^{-1}U^{\text{I},(2)\dagger},$$
$$U^{\text{I},(2)}_{p;ija} = \frac{1}{2}\sum_{bc} t^{bc}_{ij}\,\langle pa||bc\rangle - \hat{P}^{-}_{ij}\sum_{kb} t^{ba}_{ik}\,\langle pk||bj\rangle\,. \tag{D38}$$

With Eqs. (D35)–(D38) and $\Sigma(\infty)$, all third-order terms are reproduced by Eq. (D32), which completes ADC(3).

Appendix E: The EOM-IP-CCSD effective Hamiltonian

For the comparison with the effective Hamiltonian of Eqs. (71)–(73), we collect the similarity-transformed Hamiltonian of EOM-IP-CCSD in the 1h ⊕ 2h1p space.[138,139] We use spin orbitals, the antisymmetrized integrals $\langle pq||rs\rangle$, the Fock matrix f_{pq} with its off-diagonal elements (so that the expressions also hold for non-canonical and Brueckner references), the converged amplitudes t^{a}_{i} and t^{ab}_{ij}, and sum over repeated indices. The indices i, j, m, n are occupied, a, b, e, f virtual.

With $T = T_1 + T_2$, the effective Hamiltonian is

$$\bar{H} = e^{-T}\hat{H}\,e^{T} - E_{\text{CC}}\,,$$

which is diagonalized in the space of the ionization operators

$$\hat{R} = \sum_{k} r_k\,\hat{a}_k + \sum_{k<l}\sum_{b} r_{klb}\,\hat{a}^{\dagger}_{b}\hat{a}_l\hat{a}_k\,.$$

Projected onto $i<j$ and $k<l$, the eigenvalue problem reads

$$\begin{pmatrix} \bar{H}^{1\text{h},1\text{h}} & \bar{H}^{1\text{h},2\text{h}1\text{p}} \\ \bar{H}^{2\text{h}1\text{p},1\text{h}} & \bar{H}^{2\text{h}1\text{p},2\text{h}1\text{p}} \end{pmatrix}\begin{pmatrix} r_{1\text{h}} \\ r_{2\text{h}1\text{p}} \end{pmatrix} = \omega\begin{pmatrix} r_{1\text{h}} \\ r_{2\text{h}1\text{p}} \end{pmatrix}, \tag{E1}$$

with $\omega = E^{N-1}_{k} - E^{N}_{\text{CC}}$. In contrast to Eq. (71), $\bar{H}$ is not Hermitian, and its left and right eigenvectors differ. The matrix

elements of the four blocks are

$$
\begin{aligned}
\bar{H}_{i,k} &= -F_{ki} \,, \\
\bar{H}_{i,klb} &= W_{klib} + \hat{P}^-_{kl}\,\delta_{il}\,F_{kb} \,, \\
\bar{H}_{ija,k} &= W_{kaij} \,, \\
\bar{H}_{ija,klb} &= \delta_{ik}\delta_{jl}\,F_{ab} - \delta_{ab}\,\hat{P}^-_{ij}\hat{P}^-_{kl}\,\delta_{jl}\,F_{ki} + \delta_{ab}\,W_{klij} \\
&\quad + \hat{P}^-_{ij}\hat{P}^-_{kl}\,\delta_{jl}\,W_{kabi} + \sum_e t^{ae}_{ij}\,\langle kl||eb\rangle \,,
\end{aligned}
\tag{E2}
$$

with $\hat{P}^-_{kl} = 1 - \hat{P}_{kl}$ as in Eq. (72). In the 2h1p–2h1p block, F_{ab} and F_{ki} dress the three lines, W_{klij} and W_{kabi} are the dressed hole–hole and electron–hole interactions, and the last term is a separable three-body term, which annihilates the pair kl with the particle b and recreates ij and a through one T_2. For $T = 0$ and canonical orbitals, $\bar{H}_{i,k} = -\epsilon_i\delta_{ik}$, the couplings are the bare $\langle kl||ib\rangle$ and $\langle ka||ij\rangle$, and the 2h1p–2h1p block is $K^1 + C^1$ of Eq. (D37); its eigenvalues are the poles of Eq. (D32) at $\omega = -(K^1 + C^1)$, since ω is an ionization energy here.

The intermediates are

$$
\begin{aligned}
\tau^{ab}_{ij} &= t^{ab}_{ij} + t^a_i t^b_j - t^b_i t^a_j \,, \\
\tilde{\tau}^{ab}_{ij} &= t^{ab}_{ij} + \frac{1}{2}(t^a_i t^b_j - t^b_i t^a_j) \,, \\
F_{me} &= f_{me} + t^f_n\,\langle mn||ef\rangle \,, \\
F_{mi} &= f_{mi} + \frac{1}{2} f_{me} t^e_i + t^e_n\,\langle mn||ie\rangle \\
&\quad + \frac{1}{2}\,\tilde{\tau}^{ef}_{in}\,\langle mn||ef\rangle + \frac{1}{2}\,t^e_i\,F_{me} \,, \\
F_{ae} &= f_{ae} - \frac{1}{2} f_{me} t^a_m + t^f_m\,\langle am||ef\rangle \\
&\quad - \frac{1}{2}\,\tilde{\tau}^{af}_{mn}\,\langle mn||ef\rangle - \frac{1}{2}\,t^a_m\,F_{me} \,, \\
W_{mnij} &= \langle mn||ij\rangle + \hat{P}^-_{ij}\,t^e_j\,\langle mn||ie\rangle + \frac{1}{2}\,\tau^{ef}_{ij}\,\langle mn||ef\rangle \,, \\
W_{mbej} &= \langle mb||ej\rangle + t^f_j\,\langle mb||ef\rangle - t^b_n\,\langle mn||ej\rangle \\
&\quad - (t^{fb}_{jn} + t^f_j t^b_n)\,\langle mn||ef\rangle \,, \\
W_{mnie} &= \langle mn||ie\rangle + t^f_i\,\langle mn||fe\rangle \,, \\
W_{mbij} &= \langle mb||ij\rangle - F_{me}\,t^{be}_{ij} - t^b_n\,W_{mnij} \\
&\quad + \frac{1}{2}\,\langle mb||ef\rangle\,\tau^{ef}_{ij} + \hat{P}^-_{ij}\,\langle mn||ie\rangle\,t^{be}_{jn} \\
&\quad + \hat{P}^-_{ij}\,t^e_i\,\langle mb||ej\rangle \\
&\quad - \hat{P}^-_{ij}\,t^e_i\,t^{bf}_{nj}\,\langle mn||ef\rangle \,,
\end{aligned}
\tag{E3}
$$

In the CCD limit, $t^a_i = 0$, they reduce to

$$
\begin{aligned}
F_{mi} &= f_{mi} + \frac{1}{2}\,t^{ef}_{in}\,\langle mn||ef\rangle \,, \\
F_{ae} &= f_{ae} - \frac{1}{2}\,t^{af}_{mn}\,\langle mn||ef\rangle \,, \\
F_{me} &= f_{me} \,, \\
W_{mnij} &= \langle mn||ij\rangle + \frac{1}{2}\,t^{ef}_{ij}\,\langle mn||ef\rangle \,, \\
W_{mbej} &= \langle mb||ej\rangle - t^{fb}_{jn}\,\langle mn||ef\rangle \,, \\
W_{mnie} &= \langle mn||ie\rangle \,, \\
W_{mbij} &= \langle mb||ij\rangle - f_{me}\,t^{be}_{ij} + \frac{1}{2}\,\langle mb||ef\rangle\,t^{ef}_{ij} \\
&\quad + \hat{P}^-_{ij}\,\langle mn||ie\rangle\,t^{be}_{jn} \,.
\end{aligned}
\tag{E4}
$$

The terms of W_{mnij} and W_{mbej} linear in T_2 are those of the pair dressings of Eq. (73) with the CCD amplitudes, in their block-I form (hh in place of pp). The line dressings $F_{mi} - f_{mi}$ and $F_{ae} - f_{ae}$ have no counterpart of this kind: a line i (or a) fluctuates there with a partner n (or m) summed over all orbitals, while in the three-particle space of Eq. (71) the partner is one of the other two lines of the configuration.

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TABLE VI. IPs (eV) of the set of Marie, Romaniello, and Loos [99] in the aug-cc-pVTZ basis, frozen core (He core of B–Ne, Ne core of Na–Ar, Li and Be all-electron). ADC(3), CCD-ADC(3) and spin-summed doubles-only EN-ADC(3) as in Ref. 29; all ADC variants and FLEX with the MP2-relaxed static self-energy. Faddeev-ADC(3) and FLEX with RPA pair channels, with the triplet particle-hole channel at first order (T1) or switched off (T0); dRPA: direct-RPA eh channels; CCD: one CCD T_2 in every pair channel. PSD1 and PSD2 on bare TDHF phonons; D and X are the direct and second-order-exchange parts of GF2. CCSD: EOM-IP-CCSD, frozen core, from the SupInf reference data (this work's own EOM-IP-CCSD agrees to 7 meV where the roots are unambiguous). GW (G_0W_0@HF, without the MP2-relaxed static self-energy), this work; EOM-CCD on the CCD of CCD-Faddeev-ADC(3). – : no sCI value, method not defined (unstable TDHF channel), or no quasiparticle. Static self-energy in the row below the header: MP2r, MP2 density, relaxed; /CCD and /EN, the MP2 functional on the density of the CCD or EN-dressed amplitudes; –, none. MAD and MSE are over the 52 states with a value for every method (C_2, BN and F_2 excluded). All calculations have been performed in the diagonal one-shot approximation.

					Faddeev-ADC(3)					FLEX				PSD1		PSD2						
IP	sCI	ADC(3)	CCD-ADC(3)	EN-ADC(3)	RPA	T1	T0	CCD	dRPA	RPA	T1	T0	CCD	bare	$-D/4$ $-X$	bare	$-$GF2	T1 $-$GF2	CCSD	EOM-CCD	GW	GF2
static		MP2r	MP2r/CCD	MP2r/EN	MP2r	MP2r	MP2r	MP2r/CCD	MP2r	MP2r	MP2r	MP2r	MP2r	MP2r	MP2r	MP2r	MP2r	MP2r	–	–	–	MP2r
Ar (1)	15.61	15.73	15.74	15.61	16.35	15.90	15.72	15.71	15.75	15.96	15.52	15.78	15.44	15.63	15.65	15.30	15.84	15.41	15.67	15.70	15.74	15.22
Ar (2)	29.19	29.76	29.78	29.28	29.34	29.38	32.67	29.26	33.06	25.36	25.47	32.57	25.34	32.23	30.92	23.49	24.88	24.97	29.88	29.91	31.12	30.36
BeO (1)	9.97	10.67	10.36	9.85	13.43[a]	10.57	10.63	10.11	10.72	7.08[a], [c]	7.37	10.14	7.05	9.15	9.25	6.42[c]	6.89[c]	6.99	9.88	9.52	9.72	8.66
BeO (2)	11.06	12.17	11.78	11.04	13.75[a]	11.75	12.01	11.42	12.25	8.06[a]	8.11	11.49	7.91	9.75	10.07	19.25[c]	7.57	7.41	11.01	10.62	10.99	9.13
BF (1)	11.15	11.18	11.30	11.11	18.61[a]	11.65	10.86	11.41	10.87	17.02[a]	11.46	11.08	12.04	11.04	11.17	14.96	16.50	11.52	11.25	11.32	11.32	10.70
BF (2)	18.61	19.21	19.07	18.72	21.45[a]	19.07	19.34	18.85	19.67	17.11[a], [c]	16.96	19.35	16.84	18.24	18.36	17.03[c]	17.08[c]	16.37	18.17	18.11	18.74	17.52
BF (3)	20.91	21.58	21.47	21.14	42.32[b]	21.33	22.24	22.01	22.64	19.54[a], [c]	19.79	22.12	19.63	20.72	21.01	17.08	18.99	19.06	21.25	21.11	21.57	20.26
BH_3 (1)	13.31	13.33	13.37	13.25	14.19	13.52	13.28	13.36	13.42	13.83	13.23	13.28	13.28	13.03	13.24	13.51	13.77	13.19	13.34	13.38	13.67	13.07
BH_3 (2)	18.31	18.35	18.42	18.23	19.10	18.50	18.69	18.36	18.85	18.24	17.76	18.70	17.79	18.59	18.48	17.64	18.09	17.62	18.40	18.44	18.69	18.28
BN (1)	11.98	12.49	12.38	11.83	–	–	–	12.94	12.21	–	–	–	9.37[c]	–	–	–	–	–	11.97	11.79	11.74	11.32
BN (2)	13.73	13.90	14.01	13.05	–	–	–	14.01	12.96	–	–	–	15.24[c]	–	–	–	–	–	13.70	13.25	13.45	11.47
C_2 (1)	12.46	12.18	12.51	12.60	–	–	–	13.21	11.55	–	–	–	14.46	–	–	–	–	–	12.98	13.22	12.92	11.98
CH_2O (1)	10.90	11.23	11.18	10.81	11.40	11.07	11.19	10.90	11.46	9.38	9.39	11.10	9.32	10.10	10.43	7.87	8.99	8.94	10.85	10.79	11.37	9.81
CH_2O (2)	14.58	14.67	14.69	14.44	16.70	14.96	14.51	14.63	14.55	14.05[c]	13.67	14.48	13.34	14.19	14.36	12.60	13.49[c]	13.62	14.61	14.50	14.58	13.88
CH_2O (3)	16.11	16.64	16.57	16.10	18.16	16.42	16.71	16.25	17.02	14.51	14.33	16.64	14.20	15.41	15.77	13.01	14.02	13.98	16.11	16.02	16.60	14.98
CH_2O (4)	17.11	17.39	17.38	17.14	22.89	17.17	17.82	17.55	18.18	14.84[c]	14.53	16.77	14.38	16.08	16.26	13.99[c]	14.04[c]	14.05	17.49	17.58	18.01	16.42
CH_4 (1)	14.38	14.48	14.52	14.32	15.20	14.59	14.37	14.45	14.51	14.68	14.12	14.37	14.14	13.94	14.20	13.96	14.52	13.98	14.39	14.45	14.75	13.84
CH_4 (2)	23.15	23.41	23.49	23.12	23.65	23.34	24.39	23.25	24.66	21.79	21.61	24.39	21.57	24.09	23.62	20.26	21.39	21.20	23.38	23.46	23.89	23.17
CO (1)	13.93	14.10	14.18	14.03	15.84	14.22	14.11	13.93	14.21	14.78	13.42	14.26	13.44	13.89	14.01	13.97	14.60	13.26	14.19	14.40	14.77	13.36
CO (2)	16.97	17.12	17.06	16.90	18.72	17.34	17.16	16.99	17.16	16.42	15.93	17.10	15.57	16.75	16.84	15.14	16.31	15.79	17.02	16.93	17.07	16.31
CO (3)	19.64	20.34	20.19	19.71	19.38	19.95	20.78	19.60	21.13	17.30	17.37	20.68	17.20	19.29	19.50	15.91	16.73	16.73	19.79	19.69	20.30	18.56
CO_2 (1)	13.73	13.93	13.90	13.73	14.84	13.95	14.01	13.65	14.11	13.11	12.63	14.07	12.32	13.40	13.59	11.97	12.84	12.34	13.79	13.81	14.21	12.79
CO_2 (2)	17.51	18.09	18.08	17.79	18.18	17.91	18.33	17.54	18.54	15.26	15.50	18.06	15.18	17.36	17.43	14.19	15.03	15.21	18.00	18.06	18.51	16.78
CO_2 (3)	18.00	18.72	18.62	18.24	18.68	18.40	18.80	18.19	19.11	16.57	16.49	18.90	16.29	17.55	17.88	14.22	15.96	15.84	18.17	18.20	18.79	16.86
CO_2 (4)	19.25	20.14	20.02	19.58	19.99	19.72	20.29	19.52	20.66	17.45	17.37	20.26	17.22	18.86	19.11	14.97	16.81	16.69	19.52	19.55	20.16	18.07
CS (1)	11.30	11.55	11.59	11.34	16.02[a]	11.50	11.56	11.06	11.60	11.50[a], [c]	9.97	11.63	9.84	10.99	11.24	10.96[c]	11.38[c]	9.79	11.55	11.83	12.38	10.40
CS (2)	12.88	12.81	12.84	12.82	18.36[a]	13.40	12.70	13.00	12.46	17.61[a], [c]	13.49	12.82	13.15	13.05	13.11	16.79[c]	17.22[c]	13.68	13.00	13.01	12.92	12.60
CS (3)	17.92	18.14	18.05	17.76	19.87[a]	17.81	18.02	17.53	18.13	19.10[a]	18.93[c]	18.63	18.62[c]	18.21	18.17	18.81[c]	19.06[c]	18.70[c]	17.22	17.29	17.99	16.63
F_2 (1)	15.73	16.23	16.13	15.59	–	15.78	16.45	15.66	17.00	–	13.35	16.32	13.28	–	–	–	–	–	15.62	15.55	16.33	14.15
F_2 (2)	18.87	19.43	19.30	18.67	–	18.59	20.02	18.16	20.66	–	14.45	18.88	14.34	–	–	–	–	–	18.95	18.87	19.90	17.27
F_2 (3)	21.10	21.30	21.40	21.08	–	21.97	20.55	21.95	20.82	–	21.88	20.33	22.39	–	–	–	–	–	21.17	21.01	20.83	20.35
H_2O (1)	12.68	13.08	13.00	12.65	13.23	12.93	13.10	12.75	13.33	11.20	11.19	13.14	11.05	12.18	12.29	9.23	10.65	10.61	12.59	12.53	12.88	11.49
H_2O (2)	14.91	15.34	15.27	14.89	15.67	15.22	15.30	15.04	15.52	13.92	13.72	15.36	13.62	14.43	14.61	12.13	13.42	13.21	14.82	14.75	15.10	13.84
H_2O (3)	19.02	19.38	19.34	19.00	20.03	19.39	19.20	19.22	19.38	18.73	18.33	19.14	18.28	18.37	18.68	17.48	18.39	17.98	18.97	18.90	19.12	18.20
H_2S (1)	10.39	10.43	10.46	10.33	11.52	10.66	10.30	10.45	10.28	11.08	10.32	10.39	10.29	10.32	10.37	10.79	11.04	10.31	10.42	10.47	10.52	9.93
H_2S (2)	13.41	13.47	13.51	13.34	14.73	13.69	13.37	13.49	13.40	14.20	13.28	13.43	13.29	13.22	13.36	13.77	14.12	13.23	13.44	13.48	13.63	12.94
H_2S (3)	15.63	15.70	15.73	15.57	16.58	15.88	15.67	15.71	15.74	16.53	15.32	15.67	15.26	15.41	15.58	15.35	16.47	15.27	15.68	15.73	15.94	15.26
HCl (1)	12.68	12.76	12.78	12.63	13.59	12.96	12.70	12.75	12.71	13.12	12.54	12.76	12.48	12.62	12.66	12.60	13.02	12.46	12.71	12.74	12.80	12.21
HCl (2)	16.66	16.76	16.78	16.62	17.75	16.95	16.70	16.76	16.73	17.35	16.59	16.74	16.56	16.54	16.66	16.84	17.26	16.52	16.71	16.74	16.89	16.25
HF (1)	16.15	16.70	16.59	16.15	16.65	16.48	16.80	16.33	17.13	14.56	14.62	16.84	14.50	15.57	15.74	11.81	13.78	13.81	16.02	15.87	16.23	14.86
HF (2)	20.05	20.51	20.42	20.04	20.93	20.44	20.46	20.25	20.69	19.31	19.10	20.50	18.99	19.58	19.80	17.46	18.80	18.57	19.95	19.80	20.07	19.10
LiCl (1)	9.90	10.07	10.07	9.88	10.75	10.23	10.07	10.01	10.09	9.88	9.50	10.11	9.39	9.86	9.89	9.23	9.72	9.35	9.96	9.96[c]	10.01	9.44
LiCl (2)	10.58	10.76	10.77	10.56	11.45	10.92	10.78	10.71	10.81	10.61	10.20	10.79	10.09	10.47	10.54	9.87	10.43	10.02	10.64	10.64	10.71	10.10
LiF (1)	11.47	12.26	12.05	11.56	12.21	12.01	12.36	11.81	12.65	9.35	9.60	12.37	9.45	10.95	11.07	6.80	8.58	8.78	11.40	11.14	11.38	10.16
LiF (2)	11.93	12.87	12.64	12.07	12.73	12.56	12.93	12.38	13.28	9.97	10.14	12.85	10.01	11.20	11.46	7.07	9.09	9.21	11.87	11.61	11.91	10.56
N_2 (1)	15.49	15.78	15.75	15.54	15.82[a]	15.61	15.84	15.37	16.02	14.22[a]	14.13	16.00	13.92	15.14	15.34	12.61	13.83	13.70	15.64	15.82	16.34	14.40
N_2 (2)	16.93	16.89	17.05	16.97	17.72[a]	17.64	16.67	17.28	16.42	15.62[a]	16.03	16.77	15.84	16.95	17.10	14.10	15.25	15.58	17.23	17.31	17.08	16.61
N_2 (3)	18.69	19.18	19.09	18.76	24.81[a]	18.75	19.67	18.41	19.95	22.58[a], [c]	18.06	19.67	17.80	18.66	18.68	22.13[c]	23.60	18.27	18.93	19.08	19.87	17.70
Ne (1)	21.46	21.99	21.89	21.49	21.90	21.82	22.18	21.70	22.54	20.30	20.36	22.24	20.27	21.03	21.17	17.13	19.44	19.48	21.33	21.16	21.43	20.35
Ne (2)	48.31	49.22	49.10	48.57	48.91	48.91	50.24	48.84	50.77	47.69	47.66	50.55	47.61	49.27	48.54	44.05[c]	46.43	46.37	48.43	48.25	47.98	47.44
NH_3 (1)	10.90	11.13	11.11	10.86	11.56	11.10	11.10	10.91	11.23	10.05	9.86	11.19	9.73	10.62	10.69	8.85	9.73	9.52	10.86	10.88	11.19	10.00
NH_3 (2)	16.60	16.85	16.85	16.56	17.52	16.89	16.69	16.74	16.86	16.54	16.06	16.64	16.05	16.00	16.30	15.52	16.27	15.80	16.59	16.59	16.86	15.88
NH_3 (3)	26.78	27.56	27.60	27.23	29.34[b]	27.24	29.46	26.93	29.82	26.30[c]	25.48[c]	29.28[c]	25.39[c]	28.99	27.80[c]	22.86[c]	25.28[c]	25.33[c]	27.86	27.88	28.44	27.69
PH_3 (1)	10.60	10.62	10.66	10.52	12.30	10.87	10.42	10.67	10.43	11.88	10.65	10.51	10.74	10.39	10.54	11.68	11.86	10.67	10.62	10.68	10.81	10.15
PH_3 (2)	13.74	13.78	13.82	13.66	14.66	13.96	13.79	13.79	13.88	14.10	13.52	13.81	13.51	13.55	13.72	13.70	14.03	13.46	13.78	13.84	14.11	13.43
PH_3 (3)	19.51	19.56	19.65	19.33	18.89[c]	19.31	22.22	18.94[c]	22.46	16.98[c]	17.18[c]	22.90	16.97[c]	22.77[c]	20.81[c]	16.48[c]	16.86[c]	17.04[c]	20.36	20.45	21.58	20.98
SiH_4 (1)	12.79	12.83	12.86	12.74	13.62	13.01	12.88	12.83	13.00	13.15	12.65	12.89	12.67	12.65	12.83	12.82	13.10	12.61	12.84	12.89	13.23	12.64
SiH_4 (2)	18.24	18.31	18.40	18.15	18.73	18.40	19.27	18.25	19.43	17.56	17.32	19.25	17.28	19.23	18.91	16.98	17.42	17.17	18.48	18.54	19.02	18.71
MAD		0.34	0.31	0.09	1.94	0.33	0.61	0.18	0.80	1.36	1.10	0.60	1.21	0.54	0.29	2.37	1.65	1.43	0.18	0.25	0.45	0.79
MSE		+0.34	+0.31	+0.01	+1.91	+0.31	+0.57	+0.11	+0.74	−0.45	−1.02	+0.57	−1.13	−0.07	−0.04	−1.51	−0.79	−1.35	+0.10	+0.09	+0.42	−0.63

[a] RPA for BF, BeO, CS and N_2 (Faddeev-ADC(3) and FLEX): roots in energy order, not assigned states, because a root of the TDHF-unstable-adjacent channel is missing or has no quasiparticle character for a deeper orbital.
[b] Not converged within the Davidson iteration cap.
[c] Pole strength $Z < 0.3$: the method has no clean quasiparticle line for this state and the entry is its root of largest Z (FLEX, PSD, GF2: within 8 eV of the ADC(3) value of the same orbital; Z from the Dyson matrices of the ADC variants).

TABLE VII. IPs (eV) of the set of Marie, Romaniello, and Loos [99] in the aug-cc-pVTZ basis with the frozen core of the sCI references: sCI reference, Faddeev-ADC(4) with CCD pair amplitudes and its pole strength Z, and the errors with respect to sCI/aug-cc-pVTZ of Faddeev-ADC(4), diagrammatic ADC(4), ADC(3) and CCD-Faddeev-ADC(3) (all aug-cc-pVTZ, frozen core). Static self-energy (row below the header): MP3u, MP3 density, unrelaxed; MP2u, MP2 density, unrelaxed; MP2u/CCD, the MP2-unrelaxed functional on the density of the CCD amplitudes. –: no value. Statistics in Table V.

		Faddeev-ADC(4)		error			
IP	sCI	IP	Z	Faddeev-ADC(4)	ADC(4)	ADC(3)	CCD-Faddeev-ADC(3)
static		MP3u		MP3u	MP3u	MP2u	MP2u/CCD
Ar (1)	15.61	15.58	0.92	−0.03	−0.03	+0.08	+0.06
Ar (2)	29.19	29.12	0.64	−0.07	+0.03	+0.54	+0.04
BF (1)	11.15	11.42	0.88	+0.27	+0.03	−0.05	+0.17
BF (2)	18.61	18.35	0.49	−0.27	−0.40	+0.79	+0.38
BF (3)	20.91	20.10	0.66	−0.81	−0.80	+1.77	+1.18
BH_3 (1)	13.31	13.34	0.92	+0.03	+0.01	+0.00	+0.02
BH_3 (2)	18.31	18.35	0.87	+0.04	+0.02	+0.01	+0.01
BN (1)	11.98	11.35	0.82	−0.63	−0.83	+0.34	+0.83
BN (2)	13.73	13.73	0.57	+0.00	−0.71	+0.19	−0.24
BeO (1)	9.97	8.25	0.74	−1.72	−1.95	+1.04	+0.41
BeO (2)	11.06	8.63	0.74	−2.42	−2.73	+1.69	+0.82
C_2 (1)	12.46	13.13	0.87	+0.66	+0.55	−0.47	+0.57
CH_2O (1)	10.90	10.55	0.87	−0.35	−0.43	+0.41	+0.08
CH_2O (2)	14.58	14.45	0.86	−0.13	−0.17	+0.06	+0.01
CH_2O (3)	16.11	15.56	0.85	−0.54	−0.62	+0.59	+0.20
CH_2O (4)	17.11	16.98	0.78	−0.12	−0.16	+0.20	+0.36
CH_4 (1)	14.38	14.39	0.91	+0.01	−0.01	+0.02	−0.02
CH_4 (2)	23.15	23.13	0.79	−0.02	−0.01	+0.17	+0.01
CO (1)	13.93	14.08	0.87	+0.15	+0.10	−0.03	−0.17
CO (2)	16.97	16.81	0.88	−0.15	−0.17	+0.24	+0.07
CO (3)	19.64	18.92	0.79	−0.71	−0.83	+0.85	+0.07
CO_2 (1)	13.73	13.61	0.87	−0.13	−0.18	+0.23	−0.06
CO_2 (2)	17.51	17.29	0.79	−0.22	−0.29	+0.44	−0.09
CO_2 (3)	18.00	17.43	0.84	−0.56	−0.69	+0.73	+0.21
CO_2 (4)	19.25	18.56	0.81	−0.69	−0.83	+0.90	+0.28
CS (1)	11.30	11.56	0.80	+0.26	+0.18	+0.01	−0.44
CS (2)	12.88	13.03	0.88	+0.15	+0.17	−0.15	+0.03
CS (3)	17.92	17.83	0.46	−0.09	+0.04	+0.20	−0.39
F_2 (1)	15.73	15.46	0.87	−0.27	−0.36	+0.44	−0.14
F_2 (2)	18.87	18.54	0.81	−0.33	−0.42	+0.50	−0.76
F_2 (3)	21.10	21.11	0.86	+0.01	−0.13	+0.11	+0.76
H_2O (1)	12.68	12.42	0.89	−0.26	−0.32	+0.42	+0.09
H_2O (2)	14.91	14.64	0.89	−0.28	−0.34	+0.42	+0.12
H_2O (3)	19.02	18.80	0.90	−0.23	−0.29	+0.30	+0.15
H_2S (1)	10.39	10.40	0.91	+0.01	−0.01	+0.01	+0.02
H_2S (2)	13.41	13.42	0.89	+0.00	−0.01	+0.03	+0.04
H_2S (3)	15.63	15.63	0.89	+0.01	−0.00	+0.04	+0.04
HCl (1)	12.68	12.65	0.91	−0.03	−0.03	+0.05	+0.04
HCl (2)	16.66	16.64	0.90	−0.01	−0.02	+0.07	+0.07
HF (1)	16.15	15.67	0.90	−0.48	−0.55	+0.67	+0.29
HF (2)	20.05	19.66	0.90	−0.39	−0.46	+0.52	+0.26
LiCl (1)	9.90	9.80	0.89	−0.10	−0.10	+0.13	+0.08
LiCl (2)	10.58	10.46	0.89	−0.12	−0.12	+0.15	+0.09
LiF (1)	11.47	10.48	0.86	−0.99	−1.11	+1.01	+0.53
LiF (2)	11.93	10.75	0.85	−1.18	−1.32	+1.14	+0.63
N_2 (1)	15.49	15.59	0.89	+0.10	+0.03	+0.17	−0.23
N_2 (2)	16.93	17.06	0.90	+0.13	+0.14	−0.23	+0.17
N_2 (3)	18.69	18.79	0.80	+0.10	+0.01	+0.36	−0.39
NH_3 (1)	10.90	10.85	0.90	−0.05	−0.08	+0.16	−0.06
NH_3 (2)	16.60	16.53	0.90	−0.07	−0.11	+0.13	+0.04
NH_3 (3)	26.78	27.14	0.27	+0.36	+0.31	+0.72	+0.11
Ne (1)	21.46	21.05	0.92	−0.41	−0.45	+0.63	+0.34
Ne (2)	48.31	47.68	0.85	−0.62	−0.67	+1.02	+0.63
PH_3 (1)	10.60	10.65	0.90	+0.05	+0.01	−0.01	+0.04
PH_3 (2)	13.74	13.76	0.89	+0.02	+0.00	+0.01	+0.01
PH_3 (3)	19.51	19.48	0.41	−0.03	+0.10	+0.02	−0.59
SiH_4 (1)	12.79	12.83	0.90	+0.04	+0.02	+0.02	+0.02
SiH_4 (2)	18.24	18.30	0.76	+0.06	+0.06	+0.04	−0.03

TABLE VIII. Formal properties of self-energy approximations. DC free: no diagram counted twice. 2nd, 3rd: exact through second and third order in v. Hermitian: $\Sigma(\omega) = \Sigma^\dagger(\omega)$ for real ω. Resolvent: $\Sigma(\omega) = \Sigma^\infty + \mathcal{U}(\omega - \mathcal{H})^{-1}\mathcal{V}$ with ω-independent matrices of finite dimension. Coupled: the amplitudes of different channels are coupled.

	DC free	2nd	3rd	Hermitian	Resolvent	Coupled
Faddeev-ADC(3)	✓	✓	✓	✓	✓	×
CCD-Faddeev-ADC(3)	✓	✓	✓	✓	✓	✓
CCSD	✓	✓	✓	×	✓	✓
GW	✓	×	×	✓	✓	×
PSDII	×	×	×	✓	✓	×
Σ^{TDHF}	✓	✓	×	×	✓[a]	×
Hedin, general	✓	✓[b]	✓[b]	×	×[c]	✓[b]

[a] Eq. (82), with $\mathcal{V} \neq \mathcal{U}$.
[b] If the vertex is iterated to the corresponding order.
[c] Not for a dynamical vertex in general.